

A retrieved-context account of selective interference in intrusive and voluntary remembering

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Abstract

In the trauma-film paradigm, a visuospatial task after a later film reminder can reduce the frequency of film-related intrusions while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that involuntary and deliberate remembering depend on different memory representations. We propose instead that the dissociation can reflect differences in how retrieval is cued and controlled. We implement this account in an emotional Context Maintenance and Retrieval (eCMR) model that simulates film, reminder, and task encoding and predicts sequences of film and non-film recall events. In the model, a film reminder reinstates film-associated context before the task. Task items are then encoded in association with that context, so later film-related context can cue task competitors as well as film items. Deliberate recall is less affected when a maintained film-retrieval goal gives film candidates an additional source of support without equally supporting task competitors. In tasks like recognition, the test presents the item, so the model evaluates the item's fit to context instead of selecting among film and task candidates. This comparison can therefore make voluntary memory appear spared because of test format rather than because voluntary remembering is generally unaffected. Interpreting selective interference this way motivates better-matched tests of intrusive and voluntary remembering and situates trauma-film findings within a broader computational account of episodic retrieval.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

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Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. From a functional perspective, events with lasting significance need to remain accessible in memory so remembered details can guide later decisions (Zhou et al., 2025). At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current goals, cues, and circumstances. Intrusive memories after trauma suggest that durable memory can strain the selectivity of retrieval. In PTSD, such memories can come to mind unbidden and poorly matched to the present, causing distress and interfering with ongoing activity (Brewin et al., 2010; Ehlers & Clark, 2000; Iyadurai et al., 2019; Krans et al., 2009). Explaining how episodic memory preserves consequential events while regulating their retrieval is therefore a shared target for both clinical science and mechanistic theories of memory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event (Holmes & Bourne, 2008; James et al., 2016). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to come to mind involuntarily. In such research, intrusion frequency can be weakly related to voluntary-memory performance, such that the two measures dissociate (Holmes et al., 2004; James et al., 2016). The same encoded episode can therefore support both involuntary and deliberate access, and these modes can come apart empirically.

Visuospatial interference tasks make this dissociation experimentally sharper. In immediate post-film studies, tasks such as Tetris can reduce later film intrusions while voluntary memory for the same material remains relatively intact (Deepröse et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). Delayed variants add a later reminder before the interference task. Participants return after roughly 24 hours to 3 days, view reminder cues for the film, and then complete Tetris. These reminder-plus-Tetris procedures can also reduce subsequent

intrusions (James et al., 2015; Kessler et al., 2020). Related procedures have been extended beyond analogue trauma films to real-life and occupational trauma, including motor vehicle accidents, emergency caesarean section, refugee trauma, and health-care work during COVID-19 (Beckenstrom et al., 2026; Horsch et al., 2017; Iyadurai et al., 2018, 2023; Kanstrup et al., 2021, 2024; Ramineni et al., 2023). At the same time, recent reviews and replication attempts caution against treating the effect as settled or uniform (Asselbergs et al., 2023; McConnell et al., 2026; Varma et al., 2024; Wessel et al., 2025). This selective interference effect therefore raises a theoretical question: when and why can an intervention alter intrusion-like and voluntary access differently, even when both concern the same recent experience? Figure 1 schematizes this target as an interaction among reminder timing, task condition, and retrieval condition.

The selective interference effect has been used to argue that intrusive and voluntary remembering depend on differently vulnerable memory representations (Brewin et al., 1996; Brewin, 2014). Dual-representation accounts give this interpretation an architectural basis. Intrusive re-experiencing is linked especially to sensory-bound representations, whereas deliberate remembering depends more on contextualized episodic representations (Brewin et al., 2010; Brewin, 2014). In immediate post-film interventions, visuospatial tasks are understood to compete for working-memory resources while sensory-perceptual imagery from the film is still being maintained and consolidated (Baddeley & Andrade, 2000). Because later intrusions are assumed to arise especially from sensory-bound representations, this competition should reduce intrusions while leaving voluntary-memory performance relatively preserved (Deepröse et al., 2012; Holmes et al., 2009, 2010; Lau-Zhu et al., 2019). In delayed post-reminder interventions, the reminder is instead treated as reactivating the relevant memory before the task, allowing the interference procedure to alter later availability through reconsolidation or post-reactivation updating (James et al., 2015; Kindt et al., 2009). Across these variants, selectivity is explained as a stronger effect on the memory basis for intrusions than for voluntary remembering.

Here we argue that preserved voluntary memory does not, by itself, establish that a separate representation was spared. Selective interference can instead arise from differences in

how film memories are cued and controlled at retrieval. In the account developed below, a film reminder reinstates context associated with the film before task material is encoded. Task material then becomes linked to that context. When later remembering is guided by film-related context, task and film material can therefore compete, interfering with film recall. The task interferes less when retrieval depends less on context to access film material. Deliberate recall can be less affected when a maintained film-retrieval goal gives film candidates an additional source of support without equally supporting task competitors. Tasks like recognition can also be less affected, because the test presents a candidate film item for judgment rather than requiring film material to be brought to mind. On this account, the selective-interference effect emerges when retrieval conditions differ in how much they depend on film-related context that also cues task material.

We formalize this retrieval account in an emotional variant ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)) of the Context Maintenance and Retrieval (CMR) model ([Polyn et al., 2009a](#)). Retrieved-context models treat context as a gradually changing state that is shaped by experience, reinstated by retrieval, and used to cue later memory search ([Howard & Kahana, 2002](#); [Polyn et al., 2009a](#); [Sederberg et al., 2008](#)). In CMR, items and context are linked by bidirectional associations. Context cues candidate items, and retrieved items or items presented at test reinstate associated context ([Polyn et al., 2009a, 2009b](#)). Emotional CMR variants retain this item-context architecture while allowing emotional or source context to shape what is bound and later retrieved ([Cohen & Kahana, 2022](#); [Talmi et al., 2019](#)). These properties make CMR a principled setting for a retrieved-context account of selective interference based on context reinstatement and item-context binding.

Applied to the trauma-film paradigm, the model treats film, delay-period, task, and filler events as a continuous sequence through which context evolves. A film reminder is implemented as item-to-context reinstatement before task encoding. Later recall is implemented as context-to-item sampling from the film and non-film items supported by the current retrieval state. When that context supports both film and task material, task items compete with film recall rather

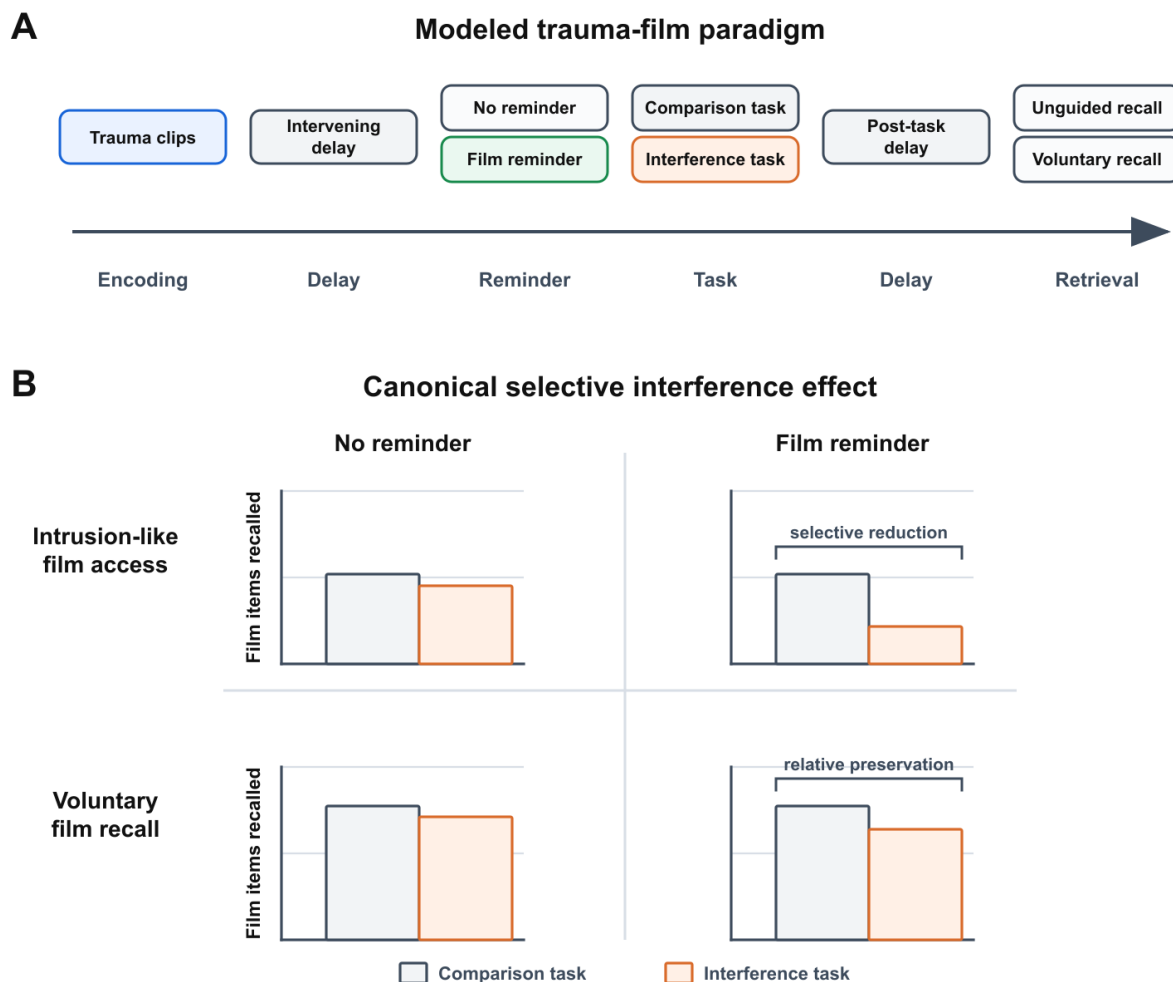
than merely weakening a film trace.

Deliberate recall is modeled with several retrieval-control processes. Rather than universally protecting film recall in a manner consistent with the selective interference effect, we show that some control mechanisms can increase film recall without making recall less sensitive to reminder-linked task encoding. In the simulations, the strongest protection comes from adding a maintained film-retrieval goal that gives film candidates support not shared by task competitors. This identifies a functional condition for selective interference: voluntary recall is protected when it depends less on film-associated temporal context that has also become supportive of task material.

The simulations follow a generalization-style modeling logic ([Busemeyer & Wang, 2000](#); [Lohnas et al., 2023](#)). We anchor core eCMR parameters in an independent affective free-recall dataset ([Dupertuys et al., 2026](#)), then map the trauma-film sequence onto that architecture and ask whether it produces the qualitative selective-interference pattern. The first simulations examine whether reminder-linked task encoding creates competitors for film recall. The next simulations ask when deliberate recall is protected from those competitors, focusing on whether a maintained film-retrieval goal gives film material a more selective basis for recall. A final recognition analysis treats recognition as a boundary condition: when a candidate film item is supplied for judgment, performance need not depend on sampling that item from a context that also cues task material. Together, the simulations define a mechanistic alternative to selective-vulnerability accounts and identify conditions under which intrusive and voluntary remembering should appear more or less dissociated.

Figure 1

Paradigm variables and canonical selective interference effect. (A) The modeled trauma-film paradigm includes film encoding, an intervening delay, a possible pre-task film reminder, an intervening task, a post-task delay, and a later test of unguided or voluntary recall. (B) The canonical selective interference effect is summarized as a schematic interaction among reminder condition, intervening task, and retrieval condition. Columns distinguish whether the task is preceded by a film reminder, rows distinguish intrusion-like film access from voluntary film recall, and paired bars compare film output after a comparison task versus an interference task. The interference task produces the largest reduction in intrusion-like film access when it follows a film reminder, whereas voluntary film recall shows a smaller reduction under the same conditions. Bars are schematic and do not plot empirical counts, fitted values, or model output.



Retrieval Conditions in Selective Interference

Figure 1 summarizes the selective interference effect as reduced intrusion-like access with relatively preserved voluntary memory for the same film material. Studies of this effect use tests that differ in whether film content must be generated or is presented for judgment. Diary measures attempt to count film memories that inadvertently arise during daily life. Laboratory intrusion tasks count memories reported during a defined monitoring period, sometimes while film cues are present. Recognition tests present candidate film statements or still images for judgment. Free recall and retrieval probe tasks require participants to produce or elaborate film details from a prompt (Holmes & Bourne, 2008; James et al., 2016; Lau-Zhu et al., 2019, 2021). The resulting comparisons differ in whether film content is already supplied or has to come to mind.

Many studies that established selective interference used recognition as the voluntary-memory test. Immediate post-film Tetris studies and delayed reminder-plus-Tetris studies report fewer diary or laboratory intrusions while verbal or visual recognition remains similar across groups (Holmes et al., 2009, 2010; James et al., 2015; Kessler et al., 2020). Later diary-recognition work uses the same basic pairing (Asselbergs et al., 2024). Those results show that the intervention can reduce film memories reported as intrusions while leaving participants able to recognize candidate film material at test. Later studies made the concern more specific: diary reports depend on film memories coming to mind, whereas recognition tests supply candidate film material for judgment (Lau-Zhu et al., 2019, 2021). Preserved recognition is therefore narrower than general preservation of voluntary memory. Reviews, replication attempts, and newer tests also caution against treating the effect as uniform (Asselbergs et al., 2023; Hemi et al., 2023; McConnell et al., 2026; Varma et al., 2024; Wessel et al., 2025).

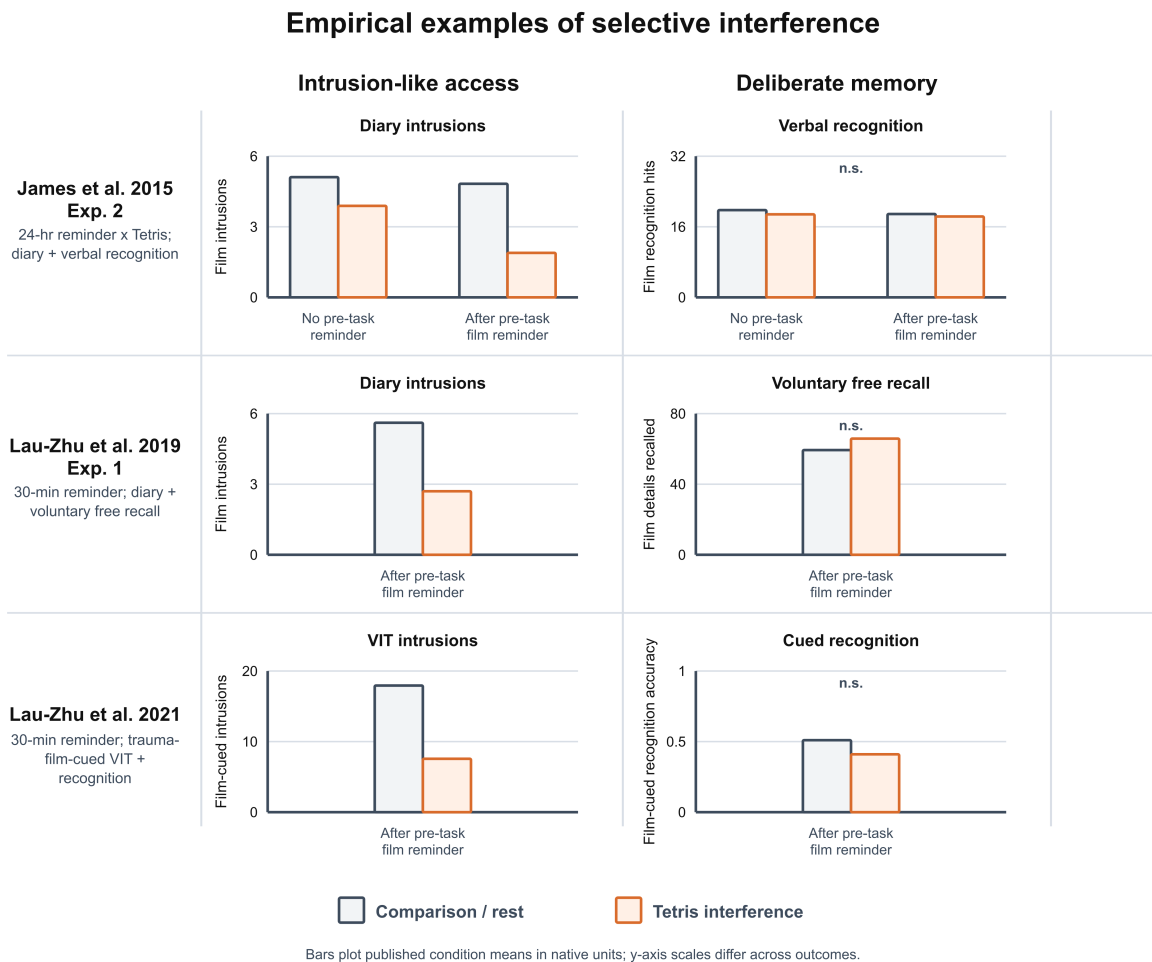
Studies using recall, probing, or vigilance-intrusion measures test whether the dissociation persists when film material has to be produced or reported under more closely matched conditions. One three-experiment study added free recall, specific probing, and priming to the diary-recognition comparison, then introduced a vigilance-intrusion task to measure intrusions during a laboratory monitoring task (Lau-Zhu et al., 2019). Reminder-plus-Tetris reduced diary

and vigilance intrusions while free recall, specific probing, and recognition were relatively preserved. Later work matched vigilance-intrusion and modified recognition measures more closely in cueing, context, timing, and attentional demands, again finding reduced intrusion measures with preserved recognition ([Lau-Zhu et al., 2021](#)). A recent preregistered study pushed the matching further by using voluntary and involuntary versions of the same vigilance retrieval task, but did not find the predicted reduction in involuntary memory ([McConnell et al., 2026](#)). Vigilance-intrusion tasks bring intrusion measurement into the laboratory. Participants perform an ongoing task and report film memories that arise, sometimes while film-related cues are presented. These cues can prompt film-related retrieval, but they do not supply the candidate film item for recognition. Vigilance-retrieval variants extend this logic by using similar task structure for voluntary and involuntary reports, changing the retrieval instruction while keeping cue timing, task context, and response format more closely matched ([McConnell et al., 2026](#)).

Figure 2 illustrates this empirical range with three exemplar designs: delayed-reminder diary-plus-recognition, delayed-reminder diary-plus-free-recall, and trauma-film-cued vigilance-intrusion-plus-recognition.

Figure 2

Empirical examples of selective interference across retrieval formats. Rows show three exemplar studies that differ in how intrusion-like and voluntary memory are measured: a delayed-reminder diary-plus-recognition design, a delayed-reminder diary-plus-free-recall design, and a trauma-film-cued vigilance-intrusion-plus-recognition design. Left panels plot intrusion-like film access and right panels plot deliberate memory. Gray bars indicate comparison or rest conditions and orange bars indicate Tetris interference. Bar heights reproduce published condition means in native study units, so y-axis scales differ across outcomes.



Panel A of Figure 1 defines the target paradigm for the trauma-film recall simulations: film encoding, a possible reminder, task encoding, and later unguided or deliberate retrieval that can

produce sequences of film and non-film recall events. The unguided retrieval condition approximates laboratory intrusion measurement rather than recognition. Film-related cues can be present during a monitoring task, but the film item still has to be sampled and reported if it comes to mind. Recognition is a boundary condition because the candidate film item is already present at test. When a test supplies the item for judgment, memory can succeed through an item-to-context route rather than through sampling from the current retrieval state. The retrieved-context account therefore has to explain why reminder-linked task material can compete with film recall during vigilance-intrusion retrieval while recognition remains relatively preserved.

A Retrieved-Context Account of Selective Interference

Figure 3

Encoding and associative structure in the retrieved-context account. (A) The model stores bidirectional item-context associations. Context-to-feature associations cue candidate items from the current context, and feature-to-context associations reinstate context when an item is recalled or when a reminder or test cue is presented. (B) The interference manipulation changes which context states task items are linked to during encoding. Without a pre-task reminder, task items are encoded after context has drifted away from the film. With a pre-task film reminder, task encoding occurs after film-associated context has been reinstated, so task items are learned in context states that overlap with film context. Thus, the same context that can cue film items can also cue task competitors during later retrieval. When task encoding is strong, those task items become stronger competitors during later context-guided retrieval. Circles mark item features: *F* labels film items and *T* labels task items. Squares mark context states. Blue marks film items or film-linked context, orange marks task items or task-linked context, green marks reminder cues, and gray marks neutral, delay, comparison, or shared model structure.

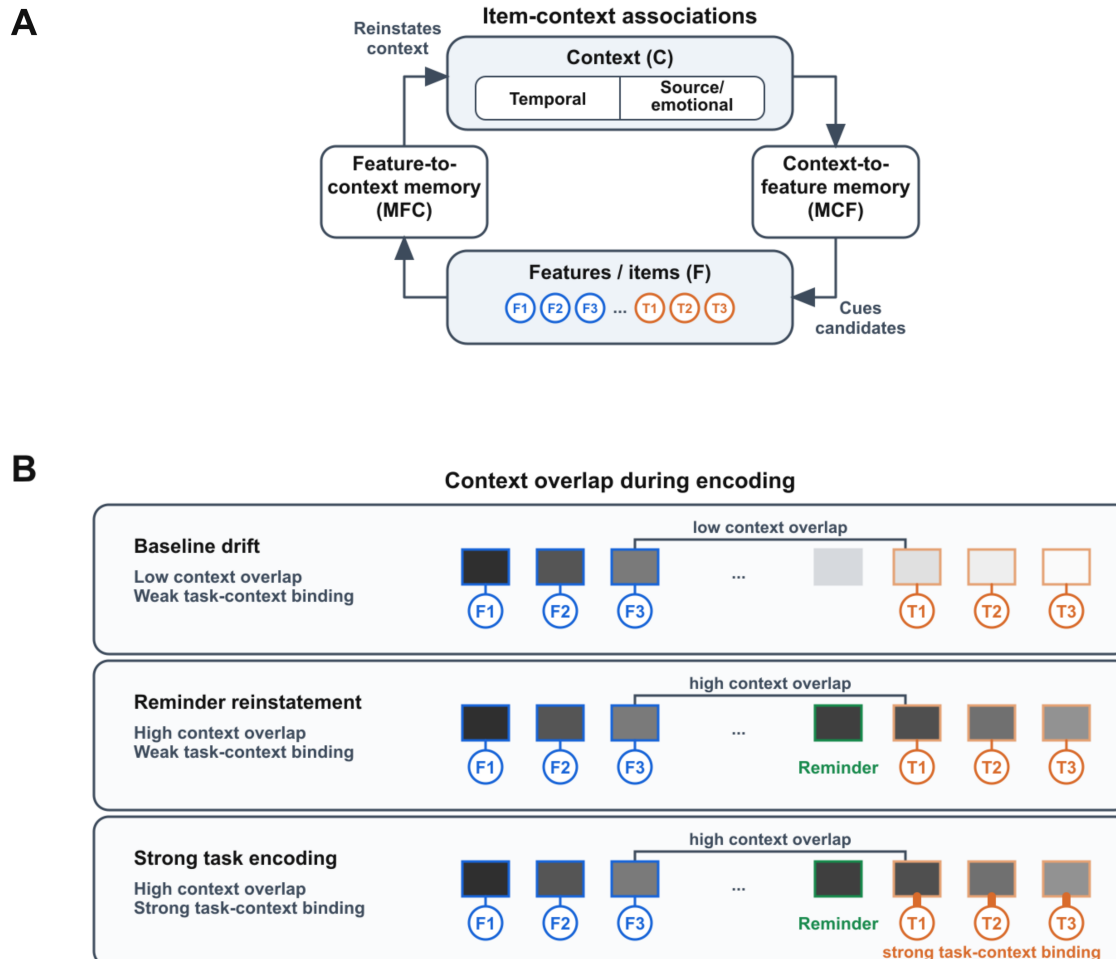
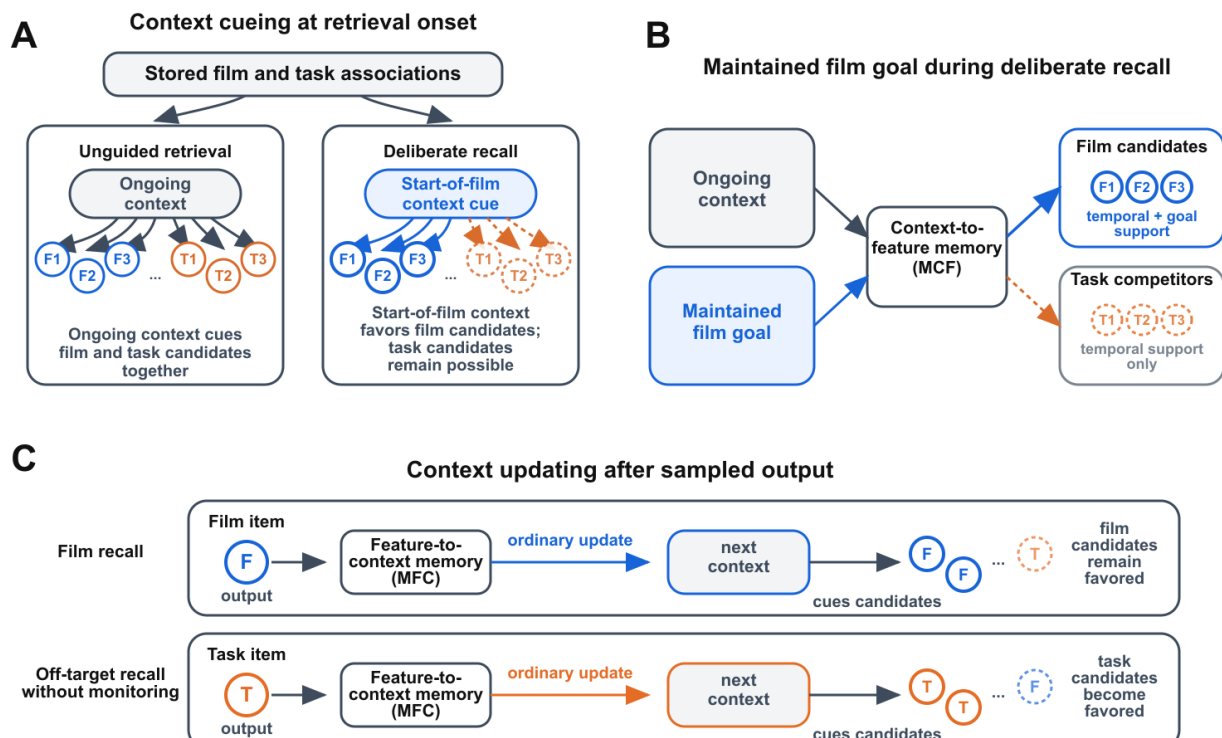


Figure 4

Retrieval operations in the retrieved-context account. (A) The upper box represents stored film and task associations; the lower boxes compare the context state used to begin unguided retrieval and deliberate recall. Stored film and task associations can be sampled from different starting contexts. Unguided retrieval begins from ongoing context, which can support film and task candidates together. Deliberate recall can begin with start-of-film context reinstatement, which favors film candidates while leaving task candidates possible. (B) A maintained film goal provides an additional deliberate-retrieval signal during context-to-feature sampling. Ongoing context can still cue both film items and task competitors, but the maintained goal adds support to film candidates rather than to task competitors. (C) Rows compare film recall, unmonitored task recall, and monitored task recall. Sampled outputs update the next retrieval context through feature-to-context memory. Recalling a film item keeps later search biased toward film candidates. Recalling a task item without monitoring can shift later context toward task candidates. With retrieval monitoring, the task-item update is dampened, preserving some film-candidate support. Blue marks film items, film-biased context, or stronger film support; orange marks task items, task-biased context, or stronger task support; gray marks shared context and associative operations. Dashed outlines or arrows mark weaker support or dampened updating, not unavailable candidates.



Retrieved-context models explain recall as search from a changing context state shaped by recent experience and prior retrieval. The current context cues candidate items, and recalled items reinstate context from encoding, producing temporal organization in free recall (Howard & Kahana, 2002; Sederberg et al., 2008). Applied to selective interference, these operations explain how task material becomes linked to film-associated context and how later retrieval determines the extent to which that material competes with film material for recall.

Film, task, and delay periods are represented as sequences of events that update context. For the film, these events correspond roughly to scenes or segments. For the task and delay periods, they represent non-film events that can also be encoded and later sampled. A film reminder reinstates film-associated context before task encoding. Task events encoded in that state can later be cued alongside film material. Retrieval then depends on how strongly the test setting constrains search toward film material rather than toward the broader set of candidates supported by the current context. Figure 3 summarizes the encoding and associative structure of the account, and Figure 4 summarizes the retrieval operations.

Context-Guided Search and Source Organization

Retrieved-context models were developed to explain structured regularities in open-ended recall. Free recall favors recently studied items, and successive recalls tend to come from nearby study positions (Howard & Kahana, 2002; Kahana, 1996; Sederberg et al., 2008). In these models, a gradually changing context state cues candidate items, and recalled items reinstate the contexts in which they were encoded. Recency follows because the retrieval context remains similar to contexts from recent study and recall events. Temporal contiguity follows because a recalled item reinstates context from its study position, which is similar to the contexts associated with nearby items.

Recall is also organized by shared content, task, and source. Semantically related items, items encoded under the same task, and items from the same source can cluster during recall (Lohnas et al., 2023; Morton & Polyn, 2016; Polyn et al., 2009a, 2009b, 2012). CMR captures this broader organization by allowing semantic, task, and source information to contribute to the

context state that guides retrieval. Items that share these forms of context can support one another during memory search even when temporal position alone would not explain the organization. In an account of the trauma-film paradigm, this lets film material receive shared film-directed support while individual film scenes or segments remain distinct encoded events.

Emotional memory gives source organization a further role in the present model. Emotional items can cluster together, and emotional material is often remembered better than neutral material (Talmi et al., 2007; Talmi, 2013; Talmi & Moscovitch, 2004). Emotional CMR represents affective information as a source-like context dimension and allows emotional items to form stronger source-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). The same architecture can therefore give trauma-film material a shared film-directed signal and stronger retrieval support, while leaving neutral task and delay material available as separate competitors.

Reminder-Linked Competitor Encoding

Reminder-linked task interventions produce interference by changing what task events become associated with during encoding. In a retrieved-context account, task events interfere with film recall when they are encoded in context states that later also support film material.

A film reminder can change the context in which the later task is encoded. Retrieved-context models assume that presented or retrieved items can reinstate context from encoding (Howard & Kahana, 2002; Polyn et al., 2009a; Sederberg et al., 2008). This operation gives film reminders their role in the selective-interference account. When a film cue is presented before the task, it shifts current context toward states associated with the earlier film episode. Task events are then encoded in a context that can later also support film material.

Recall can include material outside the intended target. In multi-list free recall, participants sometimes report items from earlier lists instead of the list they were asked to recall (Howard et al., 2008; Lohnas et al., 2015; Zaromb et al., 2006). Those errors are structured: they are biased toward recently studied prior lists and change as intervening study or delay changes. CMR explains this pattern by treating recall as candidate sampling from the current context. As context drifts, support for non-target items declines; when a non-target item was encoded in a

recent or related context, it can still enter the candidate set. This makes task and delay events eligible recall candidates in the trauma-film paradigm. If those events are encoded in context states that later support film recall, they can compete with film material during retrieval.

Delayed reminder-linked task interventions in the trauma-film paradigm instantiate this kind of competitor encoding. In these studies, participants encounter film cues before the interference task, and the procedure is typically interpreted as reactivating film memory before the task begins (James et al., 2015; Kessler et al., 2020). A reminder can also change how later material is encoded. When participants return to an earlier learning context, new material can become linked to the earlier episode (Hupbach et al., 2008; Sederberg et al., 2011). We implement the reminder as reinstatement of film-associated context before task encoding. Task events encoded in that state acquire associations with context that also supports film items. The intervention can then reduce later film recall by adding task competitors while stored film-item associations remain available. Figure 3 summarizes this encoding claim: the reminder shifts task encoding into film-overlapping context, so later context-guided retrieval can cue task competitors alongside film material.

Supplied-Item Recognition

Film-Directed Recall Control Under Matched Retrieval Conditions

Competitor encoding makes film recall vulnerable when retrieval is guided by context that supports both film and task candidates. Diary and vigilance-intrusion measures count film memories when they come to mind, often during ongoing activity or monitoring rather than after a specific film item has been supplied (Holmes & Bourne, 2008; Lau-Zhu et al., 2019, 2021). In a retrieved-context account, that kind of unguided retrieval samples from candidates supported by the current context. When task events have been encoded in film-overlapping context, the same retrieval state can support task candidates along with film candidates. Control processes can improve film recall by orienting retrieval toward the film. They protect film recall from reminder-linked interference to the extent that they make retrieval less dependent on film-related context that also supports task material.

Matched recall conditions require intrusion-like and deliberate measures to depend on context-guided candidate selection. Film cues can be present in both measures, but their role is to shape the retrieval context rather than to serve as candidate items for recognition. Free recall and specific probing move in this direction because participants must produce or elaborate film material from prompts (Lau-Zhu et al., 2019). A vigilance-intrusion design gives the simulations a closer match: film-cued laboratory tasks can hold cues, timing, and response format largely fixed while changing the instruction from monitoring for unintended film memories to deliberately searching for film memories (Lau-Zhu et al., 2019, 2021; McConnell et al., 2026). The simulated comparison therefore adapts this design to vary retrieval intentionality while holding task format and film cues fixed.

Within this matched recall setting, Figure 4 separates three ways deliberate recall can differ from unguided retrieval after the same reminder-linked task encoding. Start-of-film context reinstatement acts before sampling by changing the initial context state. A maintained film-retrieval goal acts during sampling by adding support to film candidates. Retrieval monitoring acts after sampling by reducing the carryover from off-target candidates. These control points make separable predictions because they change different parts of the retrieval cycle. The simulations test which control changes the effect of reminder-linked task encoding, rather than simply increasing film recall under all conditions.

Deliberate recall can begin from a context state biased by retrieval instructions. When people are asked to recall an earlier studied list after studying a more recent one, their early responses are often already biased toward the instructed list rather than showing a gradual search backward through recent-list items (Healey & Wahlheim, 2024; Lohnas et al., 2015). Recall also shows a reliable tendency to start near the beginning of a studied list, especially when the usual end-of-list advantage is weakened by a delay (Healey & Kahana, 2014; Herrema & Kahana, 2024; Howard & Kahana, 1999). Start-of-list context reinstatement mechanisms account for these patterns by evolving the current context toward the state associated with the beginning of the target list before retrieval begins (Healey & Wahlheim, 2024; Kragel et al., 2015; Lohnas et al.,

2015; Morton & Polyn, 2016). In our retrieval control simulations, start-of-film reinstatement drifts the retrieval state toward the context present at the beginning of film encoding. This starting state favors early film candidates before ordinary context-guided sampling resumes, so its predicted benefit is tied to the temporal region that has been reinstated.

Retrieval monitoring acts after a candidate has been sampled. Evidence for this post-sampling control comes from externalized recall tasks where participants report every word that comes to mind and mark errors. These instructions reveal many more intrusions than standard recall, and many sampled intrusions are marked as errors, indicating that off-target candidates can enter retrieval without being accepted as target memories (Lohnas et al., 2015; ?). Monitoring also carries forward across retrieval: when people fail to identify one response correctly, they are worse at rejecting the next intrusion (?). At the same time, sampled intrusions can pull search away from the target list, as intrusions tend to cluster with other intrusions from related prior-list or semantic contexts (Zaromb et al., 2006; ?). Retrieved-context implementations have treated monitoring as a target-match filter while still allowing rejected items to update the next retrieval context (Lohnas et al., 2015). The trauma-film simulations explore whether that post-sampling evaluation can also reduce carryover from sampled non-film candidates. A sampled non-film candidate still counts as a retrieval attempt, but its context update is dampened when it is evaluated as off-target. Monitoring can therefore help deliberate recall return to film material after non-film competition appears, while leaving the initial competition from reminder-linked task encoding in place.

A maintained retrieval goal gives deliberate recall a broader film-directed signal during candidate sampling. When people study mixed-category material and are asked to retrieve only one category, they can restrict recall to the target subset while temporal organization remains visible within that subset (?, ?). In this kind of task, category-selective activity tracks retrieval effort, remains active between reported recalls, and weakens before extra-category intrusions (?). Existing CMR models already allow semantic, task, and source context to shape candidate support (Morton & Polyn, 2016; Polyn et al., 2009a, 2009b). Category-cued search has been

formalized in associative memory models, and recent review work treats this as a natural but still underdeveloped direction for CMR (?; ?). The simulations therefore add a maintained film-specific cue during candidate sampling. This cue gives film candidates additional support without equally supporting task competitors. Task items encoded in film-overlapping context can still compete through temporal context, but they do not receive this goal-based support.

A retrieval-control process can increase film recall relative to unguided retrieval without protecting film recall from a reminder-linked task intervention. Protection depends on whether recall becomes less reliant on film-associated temporal context after that context has also become supportive of task material. Start-of-film reinstatement and retrieval monitoring both improve film recall in the tested simulations, but neither adds a cue that selectively supports film items over task items encoded in film-overlapping context. The maintained film-retrieval goal tests that different route. It gives film candidates support from a film-specific cue, so task competitors remain available but do not receive the same added support. Under the tested conditions, that goal-based support limits the intervention’s impact enough to simulate a selective-interference effect.

Model Specification

The formal model keeps the standard CMR separation between item features, context, feature-to-context memory, and context-to-feature memory. The section first defines these inherited representations and base parameters, then specifies the context-updating, learning, retrieval, and control operations used to configure the selective interference simulations.

Representations and Parameters

Each study event i has an orthonormal feature vector $f_i \in \mathbb{R}^N$, where N is the number of encoded item features in the simulated paradigm. The model maintains a unit-length context vector $c_i \in \mathbb{R}^D$, where D is the number of context units. The core context units represent the start state and item-associated temporal context. Before encoding begins, context is initialized to the start state c_0 , with the start unit active and item-associated temporal-context units inactive. Subsequent events update this state, so the context associated with each film, task, or filler item depends on where it occurs in the continuous paradigm stream. The model contains two

associative matrices. The matrix $M^{FC} \in \mathbb{R}^{D \times N}$ maps feature vectors to context input. The matrix $M^{CF} \in \mathbb{R}^{N \times D}$ maps context to feature support. All vectors are treated as column vectors, so $M^{FC}f$ is a context vector and $M^{CF}c$ is a feature-support vector.

Table 1 lists the model parameters used in the equations below. The notation is adapted across several retrieved-context and CMR implementations, including foundational free-recall models (Polyn et al., 2009a; Sederberg et al., 2008), later likelihood-based variants (Kragel et al., 2015; Morton & Polyn, 2016), emotional CMR variants (Cohen & Kahana, 2022; Talmi et al., 2019), and target-monitoring extensions (Lohnas et al., 2015).

Pre-experimental associations represent background memory structure already present before the simulated paradigm begins. They give each item baseline self-support and shared support, so later retrieval reflects both pre-existing structure and associations learned during the film, task, and filler phases. The selective interference simulations vary the new associations formed across phases. These baseline associations are fixed model architecture and are held constant across the selective interference simulations. For the start unit and item-associated temporal context units:

$$M_{pre}^{FC}(j, i) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$M_{pre}^{CF}(i, j) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases} \quad (2)$$

The start context has no pre-experimental context-to-feature support. For context units corresponding to items, δ gives each item self-support and α gives all other items shared support. This initialization supplies baseline support before experimental learning.

Context Updating, Learning, and Retrieval

Context updating defines both gradual drift during encoding and reinstatement by recalled items and by items presented as reminders or test cues. The same feature-to-context operation updates context when an item is encoded, recalled, presented as a reminder, or presented as a cue by the test. When feature f_i is active, it probes feature-to-context memory to retrieve context input:

$$c_i^{\text{IN}} = M^{FC} f_i \quad (3)$$

The model normalizes this input to unit length, yielding $\hat{c}_i^{\text{IN}}$. Context then drifts toward the normalized input with a drift rate β :

$$c_i = \rho_i c_{i-1} + \beta \hat{c}_i^{\text{IN}} \quad (4)$$

Let $d_i = c_{i-1} \cdot \hat{c}_i^{\text{IN}}$. The normalization coefficient is:

$$\rho_i = \sqrt{1 + \beta^2 (d_i^2 - 1)} - \beta d_i \quad (5)$$

The coefficient ρ_i is the nonnegative solution that keeps c_i at unit length. Associative learning defines how encoded film, task, and filler events become later retrieval candidates. When a new event is encoded, the active feature and updated context are associated via outer-product learning. Feature-to-context associations update as:

$$\Delta M^{FC} = \gamma c_i f_i^{\top} \quad (6)$$

A primacy function weights context-to-feature learning:

$$\phi_i = \phi_s e^{-\phi_d \cdot i} + 1 \quad (7)$$

Context-to-feature associations update as:

$$\Delta M^{CF} = \lambda \phi_i f_i c_i^{\top} \quad (8)$$

Here, λ scales context-to-feature learning, while ϕ_s and ϕ_d define the size and decay of the primacy boost. Retrieval uses the current context state to assign support to candidate items, sample from the resulting competition, and terminate the sampling sequence. During context-guided retrieval, temporal context and source context cue encoded items through their context-to-feature associations. A maintained film-retrieval goal can also contribute source/goal-like support to candidates that match the film target:

$$a_i = [M^{CF}c]_i + [M_{emot}^{CF}c_{emot}]_i + \omega_{goal}g_i \quad (9)$$

Here, c_{emot} denotes the source-context state, and g_i is the degree to which item i matches the maintained film-retrieval goal. Film items match the goal; task, break, and filler items do not.

The choice sensitivity τ controls how strongly support differences affect choice among candidates:

$$A_i = a_i^\tau \quad (10)$$

The probability of sampling item i , conditional on continuing retrieval, follows a Luce choice rule over currently unsampled items:

$$P(i \mid \text{continue}) = \frac{A_i}{\sum_k A_k} \quad (11)$$

At each retrieval step, the model may terminate with probability $P_{\text{stop}}(j)$ or sample an item with probability $(1 - P_{\text{stop}}(j)) \cdot P(i \mid \text{continue})$. Termination probability grows with retrieval-attempt position:

$$P_{\text{stop}}(j) = \theta_s e^{j\theta_r} \quad (12)$$

Here, j is the number of items already sampled, θ_s sets the baseline termination probability, and θ_r controls its growth. In simulations, this value is bounded to remain a valid probability. After a sampled item r_j , context is updated through the same feature-to-context pathway:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \beta_{\text{rec}} \widehat{M^{FC} f_{r_j}} \quad (13)$$

This sampled-item update drives context toward the sampled item’s encoding neighborhood, biasing subsequent retrievals toward nearby items and producing temporal organization in retrieval. Encoding operations use the context update and learning equations together, whereas reminders, sampled items, and film items supplied as test cues use the context update without adding a new recallable event.

Retrieval Control

Retrieval-control settings alter the starting state, goal-directed support, and context updating used by the same context-to-feature retrieval operation. Unguided retrieval and deliberate recall use the same retrieval-support and choice rules in Equation 9 and Equation 11. They differ in the control settings applied to the retrieval state and sampling process. The first deliberate-recall control setting is start-of-film context reinstatement:

$$c_{\text{start}} = \rho_{\text{start}} c_N + \beta_{\text{start}} c_0 \quad (14)$$

Here, ρ_{start} is calculated from Equation 5 using β_{start} . Simulation settings can use the fitted β_{start} value directly or multiply it to vary the degree of start-of-film context reinstatement. The second control setting is a maintained film-retrieval goal. It is represented by the goal-match term $\omega_{\text{goal}} g_i$ in Equation 9, so a film-directed goal contributes support to film candidates without giving the same support to task or filler candidates. The third control setting is retrieval monitoring after an off-target item is sampled. Let $t_i = 1$ for film items and $t_i = 0$ otherwise, and define $\kappa_i = t_i + (1 - t_i)\kappa$. Retrieval monitoring modifies the sampled-item context update in Equation 13. With retrieval monitoring, the context update after sampled item i becomes:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \kappa_i \beta_{\text{rec}} \widehat{M^{FC} f_i} \quad (15)$$

When $\kappa < 1$, sampled off-target candidates have less influence on the next context state

than sampled film items. This convention is used for deliberate recall when simulations manipulate rejected-sample updating. Recognition-like tests are not retrieval-control settings in this sense. When a test presents a candidate film item, the item is represented as cue-driven context reinstatement using the same feature-to-context input operation as Equation 3:

$$c_{\text{film}}^{\text{IN}} = M^{FC} f_{\text{film}} \quad (16)$$

Candidate film items presented by the test therefore use the same feature-to-context operation as sampled and reminded items.

Source Context

The source-context pathway adapts the emotional/source-context machinery of eCMR and CMR3 to the trauma-film setting. In published emotional CMR models, emotional features update a source-context subregion and can form item-context associations (Cohen & Kahana, 2022; Talmi et al., 2019). The simplified film-source dimension fills the same architectural role: a non-temporal context feature that can organize recall when shared across items (Cohen & Kahana, 2022; Polyn et al., 2009a; Talmi et al., 2019). This pathway is represented by a simplified source feature vector s_i with a marked film/source feature and a neutral/default feature. Film items are assigned the marked source feature, whereas break, task, and filler items are assigned the neutral/default feature. The source feature vector initializes M_{emot}^{FC} so that an item can cue its source-context components:

$$M_{\text{emot}}^{FC}(\cdot, i) = (1 - \gamma)s_i \quad (17)$$

Source-context-to-feature support begins at zero:

$$M_{\text{emot}}^{CF}(i, \cdot) = 0 \quad (18)$$

Thus, source context can update during experience before it contributes retrieval support. Source-context components use the same context-update rule in Equation 4 with source-context

drift rate β_{emot} . Let η_s denote baseline source-context-to-feature learning and let $\phi_{emot,i}$ denote added learning for item i in the marked source pathway. With the binary vector above, $\phi_{emot,i} = \phi_{emot}[s_i]_1$, so film items assigned the marked source feature receive ϕ_{emot} and neutral/default items receive 0. When source-context learning is engaged, the source components of context receive additional context-to-feature learning:

$$\Delta M_{emot}^{CF} = \lambda \phi_i (\eta_s + \phi_{emot,i}) f_i c_{emot,i}^\top \quad (19)$$

Here, $c_{emot,i}$ denotes the source components of context. Source-context support enters retrieval through learned source context-to-feature associations and contributes to the same competition defined in Equation 9. Because η_s is nonzero, neutral/default items can weakly bind the source context active during encoding without being assigned the marked film/source feature. This lets task items inherit some influence from reinstated film-source context when the task is encoded after a film reminder, while still preserving a distinction between marked film items and neutral task competitors.

Retrieval begins with fixed partial reinstatement of the marked film-source context, governed by $\beta_{film-src}$. This fixed source-context orientation represents the test setting orienting retrieval toward the film episode at the level of source context. It is held constant across unguided and deliberate retrieval and is separate from the start-of-film context reinstatement manipulated in Simulation 2.

These equations define the item-context architecture used in all simulations. During retrieval, the model samples one candidate at a time until stopping. Task competitors reduce film recall when they are sampled instead of film items, using retrieval attempts that could otherwise have returned film material.

Simulations

The simulations instantiate the retrieved-context account in a controlled trauma-film analogue: a laboratory task in which distressing film material is used to study memory processes relevant to responses to real-life trauma (Holmes & Bourne, 2008; James et al., 2016). The

simulations are organized into three linked groups: reminder-linked task competition, retrieval-control consequences for deliberate recall, and recognition-like test-format consequences of bidirectional item-context associations. Simulation 1 shows how a task performed after a film reminder attenuates film recall when task encoding is strong and occurs in context states that overlap with film encoding context. Simulation 2 adds the intentionality manipulation by comparing unguided retrieval with deliberate recall and then decomposing the control settings that protect film recall during deliberate retrieval. The recognition simulation compares context-to-item recall with a test format in which a candidate film item is presented for judgment.

Methodology

The simulations require default values for context drift, associative learning, retrieval choice, and stopping before the trauma-film manipulations are applied. Ideally, these values would be estimated from a selective interference experiment that implements the full paradigm modeled here: film encoding, reminder or no-reminder conditions, an interference task, a later retrieval test, and an observed dissociation between unguided and deliberate retrieval measures. Such a dataset would also need trial-by-trial recall sequences that identify which film, task, and non-film candidates entered the retrieval stream and in what order, so the model could estimate retrieval competition rather than only final film memory scores. Existing trauma-film studies define the target phenomenon, but they typically do not provide this model-fitting structure. We therefore use an immediate affective free-recall dataset as a provisional anchor for core eCMR dynamics before applying the selective interference manipulations in simulation. The anchor dataset included participants who reported concern about the war in Ukraine and studied and recalled lists of war-related and neutral photographs ([Dupertuys et al., 2026](#)). The fitted dataset included 123 participants, four lists per participant, 492 immediate recall sequences, and 3,804 scored recall events. The fit provides an externally estimated eCMR regime for deliberate context-guided recall of affective and neutral photographs. The fitted parameters set the starting encoding, retrieval, and stopping dynamics before the selective interference manipulations are applied as phase-specific encoding or retrieval-control settings. In these simulations, film items

reinstate the marked source feature into source context, neutral/default events can weakly bind active source context, and retrieval begins with partial film-source reinstatement. To accommodate the difference in total encoded items between the immediate-recall paradigm and these simulations, the primacy profile is also moderated relative to the immediate-recall fit so that the film sequence is not dominated by an unrealistically steep first-item advantage. Table 3 gives the default numerical values used in the simulations.

The simulations update associations after context integration and disallow repeated sampling of the same item. Across simulations, each condition sets phase-specific drift, learning, and retrieval-control parameters, then runs the model through encoding and retrieval. Film, break/buffer, task, and filler items are encoded by the same CMR operations defined above. When modeled as cue events, reminder cues and candidate film items presented by the test update context through Equation 3 and Equation 4 without adding new recallable items. Simulations that present candidate item information in a different test format state that separately. Each simulated trial contains 16 film items, 16 break/buffer items, 16 task items, and 16 filler items unless otherwise noted. Equal phase lengths make redistribution across film, task, and filler periods directly comparable. Retrieval allows up to 48 sampling attempts. In context-to-item recall simulations, all encoded film, break/buffer, task, and filler items can enter the retrieval competition. The simulated retrieval sequence records sampled candidates rather than only final accepted responses. This convention lets retrieval monitoring count an off-target sample as a retrieval attempt, attenuate its context update, and still allow search to continue.

A parameter sweep involves rerunning the full encode-then-retrieve simulation across planned values of a manipulated parameter. For each simulation condition and manipulated parameter value, the model first completes the phase sequence, then enters retrieval and samples recallable items using Equation 11 and Equation 12. This procedure is repeated across simulated trials within each condition/value cell. When figures plot recall across the encoded stream, they summarize retrieval after the full phase sequence has been encoded. Recall curves are estimated across simulated trials by scoring which encoded items appeared in each retrieval sequence.

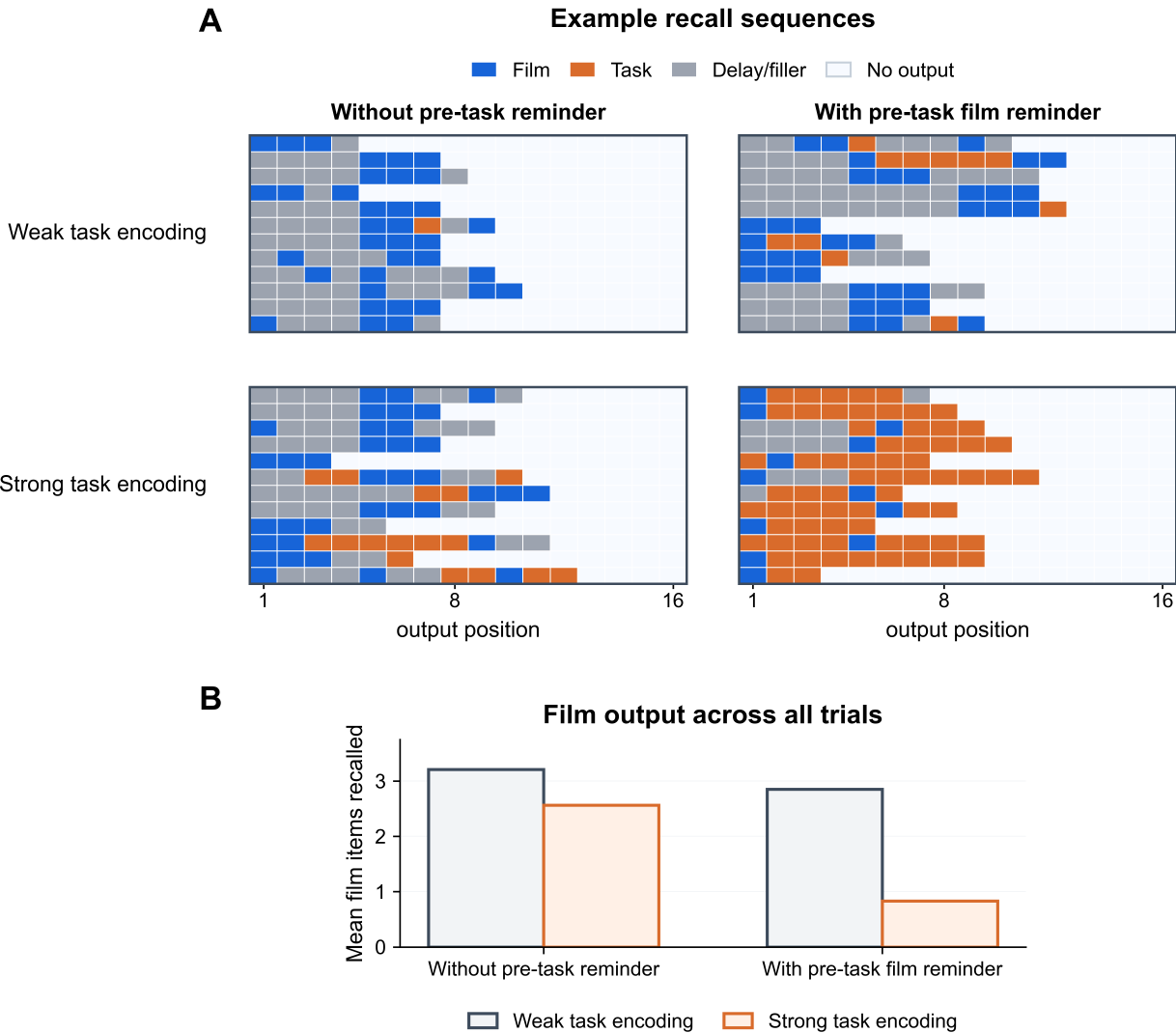
Serial-position curves plot the proportion of trials in which each encoding position was retrieved, and phase summaries average those proportions within phase. Thus, position and phase labels identify the encoding positions, whereas recall probabilities refer to subsequent retrieval.

Simulation 1: Task Encoding and Contextual Overlap

Simulation 1 demonstrates how a post-film task can reduce later film recall when task events are encoded in context states that overlap with the film sequence. To separate task encoding strength from reminder-driven context reinstatement, the simulation runs both the no-reminder and with-reminder conditions across the same set of task encoding strengths. We express task encoding strength as a task encoding multiplier that scales task-phase context-to-feature learning in Equation 8: 0 removes new task context-to-feature learning, 1 applies the baseline fitted rate, and values above 1 strengthen task item-context binding. In the no-reminder condition, the reminder-start and film-cue drift scales are set to zero. In the with-reminder condition, context first shifts toward the start state and then receives sequential film reminder cues before task encoding. Reminder cues update context without applying Equation 6 or Equation 8 and without adding new recallable items. Retrieval settings are held fixed, so differences in recall reflect the item-context associations formed before retrieval rather than changes in retrieval control. Because each film position is encoded with a different temporal-context state, the model predicts a serial-position pattern, not only an overall change in film recall. Task items should compete most with the film positions whose encoding contexts are most similar to the contexts active during task encoding. Simulation 1 first isolates the unguided side of the effect summarized in Figure 1. It implements the empirical task contrast as a manipulation of task encoding strength, crossed with whether a film reminder precedes task encoding. In the fixed-cue version shown in Figure 5, weak film cues are supplied periodically during recall in all conditions.

Figure 5

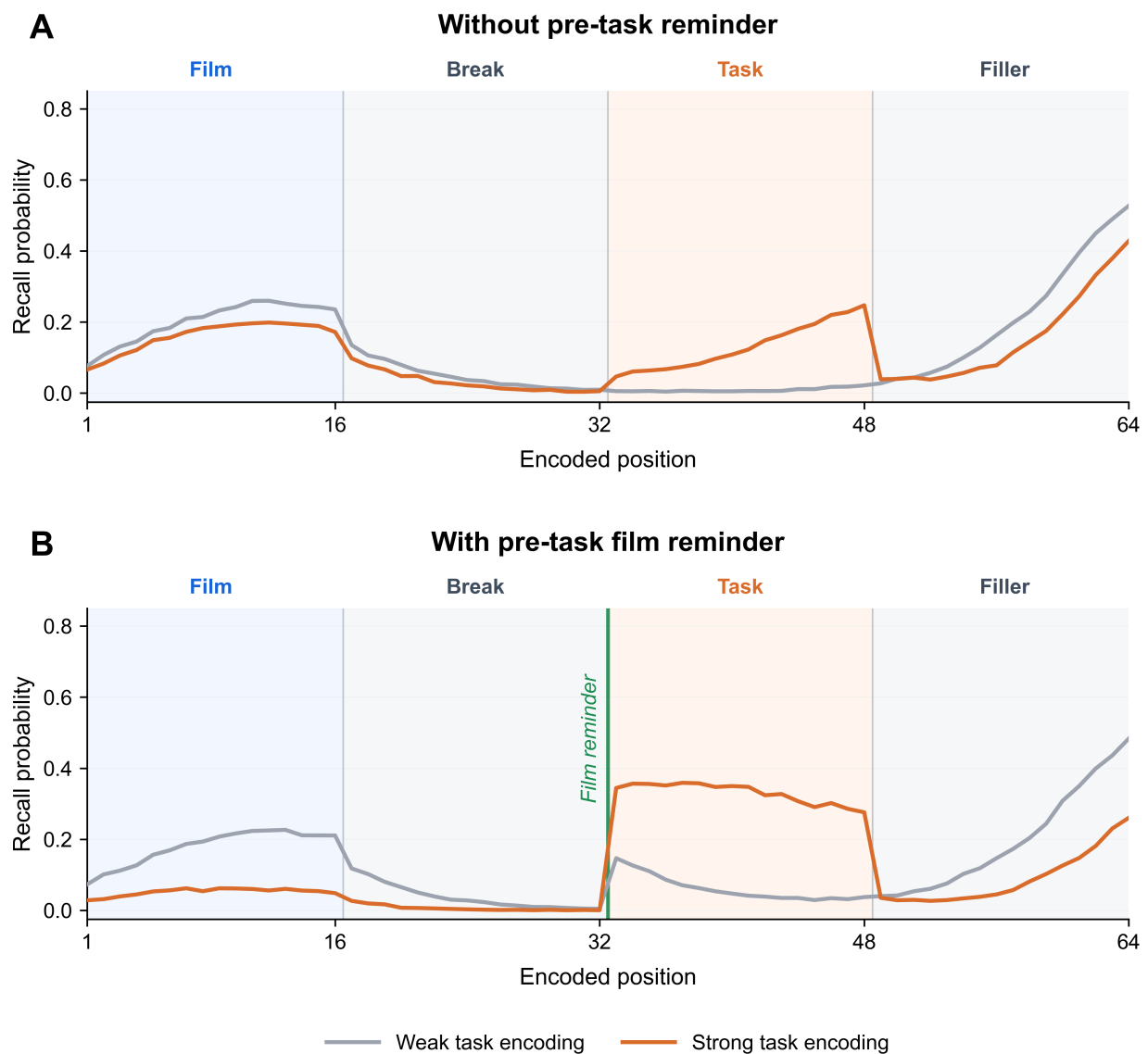
Simulation 1: unguided film recall after task encoding. Reminder condition and task encoding strength were crossed while retrieval mode was held unguided; weak film cues were supplied periodically during recall in every condition. **(A)** Example generated recall sequences for each condition. Each horizontal lane is one simulated trial selected near the condition mean for film output, with output positions running left to right. Colors mark film items, task items, delay/filler items, and unfilled output positions. **(B)** Mean number of film items recalled across all simulated trials. Film-item recall was lowest when a film reminder preceded strong task encoding, isolating the unguided-retrieval component of the selective interference effect.



These aggregate results establish the unguided interference component. The diagnostics below first show how this redistribution appears across encoded positions, then examine how it follows from context overlap; Simulation 2 then asks which deliberate-retrieval controls attenuate this loss. Table 4 summarizes the phase structure and settings used for Simulation 1. In the table, break/buffer items implement the intervening delay, and filler items implement the post-task delay. For each reminder condition and task encoding strength, serial-position curves show retrieval-test outcomes sorted by where each item appeared in the encoded sequence. Figure 7 inspects the context-overlap structure that helps explain the aggregate film and task output in Figure 5. The context-similarity analysis asks how similar each film-encoding context is to the contexts active while task items are encoded, using cosine similarity over unit-normalized temporal context states.

Figure 6

Simulation 1: reminder-linked task encoding redistributes recall probability by encoded position. Reminder condition and task encoding strength were crossed while retrieval was held unguided; weak film cues were supplied periodically during recall in all conditions. **(A)** Without a pre-task film reminder, recall probability by encoded position changes only modestly when task encoding is strengthened. **(B)** With a pre-task film reminder, strong task encoding reduces recall from film positions and increases recall from task positions. Curves compare weak and strong task encoding; shaded bands mark film, break, task, and filler positions, and the green vertical line marks the pre-task film reminder. Task encoding strength therefore changes recall probability by encoded position most clearly when task encoding follows a film reminder.



Without a pre-task film reminder, increasing the task encoding multiplier produces only modest loss of film recall and only modest gain in task recall, because task items become associated with later context states reached after drift away from the film. With a pre-task film reminder, the same task-encoding manipulation produces a larger drop in film-position recall and a larger rise in task-position recall. The condition with a pre-task film reminder therefore shows that task encoding strength matters most when task items are encoded in a film-related context state. In the selected weak-versus-strong comparison, mean film recall in the no-reminder condition changes from 3.211 to 2.562 items, while mean task recall changes from 0.143 to 2.105 items. In the condition with a pre-task film reminder, mean film recall changes from 2.852 to 0.830 items, while mean task recall changes from 0.987 to 5.282 items.

Figure 7***Simulation 1: reminder reinstatement increases overlap between film and task contexts.***

Reminder condition was varied before task encoding while retrieval was held unguided. (A) Mean task-context overlap for each encoded film position, averaging across encoded task positions. (B) Full film-by-task context-overlap matrices for the same conditions; each cell compares the context state associated with one encoded film position to the context state active at one encoded task position. Brighter cells indicate greater context overlap. The film reminder increases overlap between film-encoding and task-encoding contexts, especially for later film positions and early task positions.

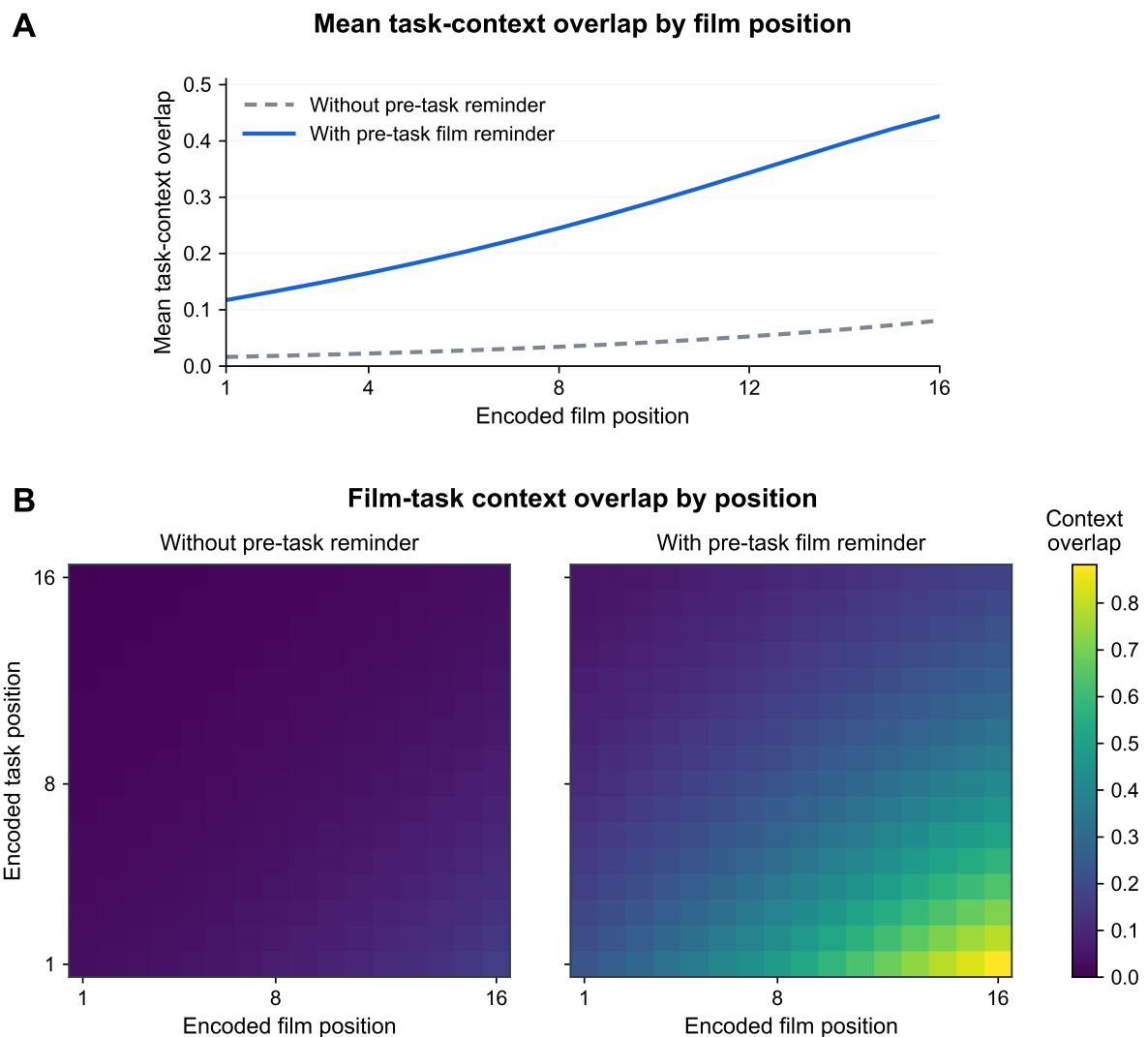


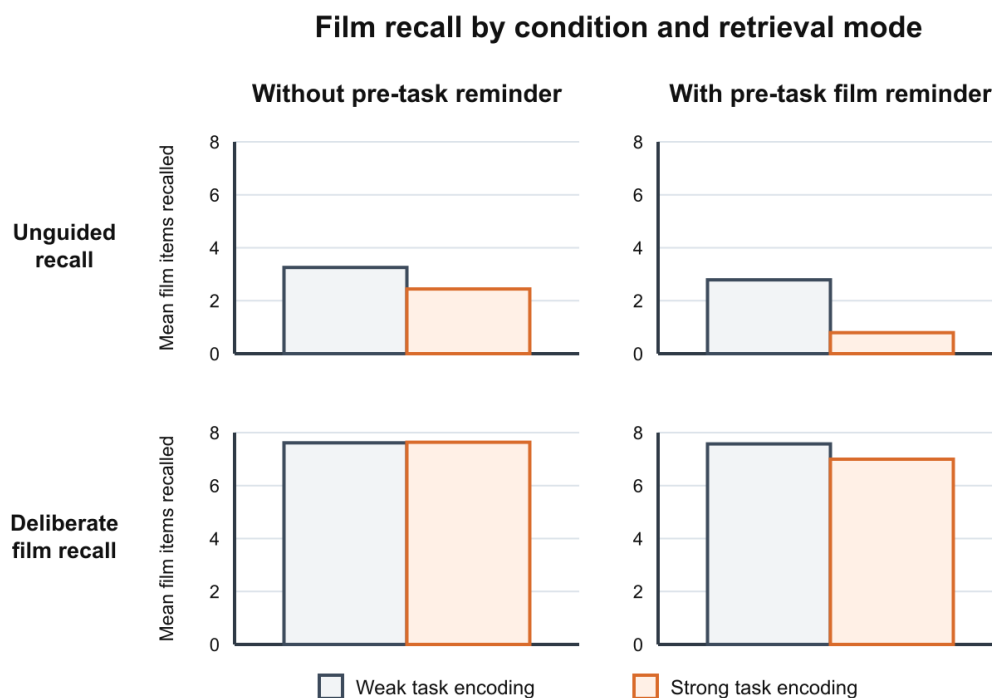
Figure 7 explains why the no-reminder and with-reminder curves diverge. Panel A summarizes temporal-context overlap between the states associated with film items and the states active during task encoding. When a sequential film reminder reinstates film-associated context before task encoding, the task-encoding contexts become more similar to some film-position contexts than to others. Panel B shows the corresponding film-position by task-position overlap matrix. Task encoding strength then determines how strongly task items are bound to these overlapping task-encoding contexts, while film encoding settings remain unchanged. The loss of film recall comes from stronger support for task items that can be cued by film-associated context. In paradigm terms, the reminder leaves the model partly oriented toward the film as task encoding begins, so task events become linked to film-like temporal context as well as the active film-source state. Task items remain distinct from film items but become increasingly likely to win retrieval from film-related cues. Simulation 2 next examines whether deliberate retrieval control reduces the impact of these task-item associations through start-of-film context reinstatement, a maintained film-retrieval goal, and retrieval monitoring.

Simulation 2: Control Mechanisms for Deliberate Recall

Simulation 2 turns from encoding conditions to retrieval control. It examines when deliberate film recall remains protected as reminder condition and task encoding strength are crossed. The deliberate-recall setting combines three control mechanisms: start-of-film context reinstatement, a maintained film-retrieval goal, and retrieval monitoring after off-target samples. Start-of-film context reinstatement drifts context toward the state present at the start of film encoding before the first retrieval attempt. The maintained film-retrieval goal adds support for film items during candidate sampling. Retrieval monitoring dampens context updating after off-target samples; in the anchor simulation, off-target context updates are partially dampened rather than fully suppressed. The figure compares this deliberate-recall setting with unguided recall across the full reminder-by-task-encoding design.

Figure 8***Simulation 2: deliberate film recall is protected from reminder-linked task interference.***

Reminder condition and task encoding strength were crossed as in Simulation 1 while retrieval mode was varied between unguided recall and deliberate film recall. Deliberate film recall combined start-of-film context reinstatement, a maintained film-retrieval goal, and moderate retrieval monitoring. Weak film cues were supplied periodically during recall in all conditions. Rows compare retrieval modes, columns compare reminder conditions, and paired bars compare weak versus strong task encoding. Strong task encoding reduced film recall most clearly when it followed a film reminder, but this reduction was much larger under unguided recall than under deliberate film recall. The model therefore simulates a selective interference effect: the reminder-plus-strong-task intervention strongly reduces unguided film access while leaving deliberate film recall relatively preserved.



In unguided recall, strengthening task encoding reduced film output modestly without a film reminder and strongly after a film reminder. In deliberate film recall, the corresponding

reduction after a film reminder was much smaller. The next figure decomposes this deliberate-retrieval setting to show which control ingredients are responsible for preserving film recall under the high-interference endpoint.

Figure 9

***Simulation 2: different retrieval controls preserve film recall to different degrees.** Each control setting was evaluated at two endpoints of the interference manipulation: one without a pre-task reminder and with weak task encoding, and one with a pre-task film reminder and with strong task encoding. Weak film cues were supplied periodically during recall in all conditions. Mean film-item recall is shown for each control setting at the two endpoints. Start-of-film reinstatement and retrieval monitoring increase film recall in some conditions, but the smallest endpoint differences occur when deliberate recall includes a film-retrieval goal.*

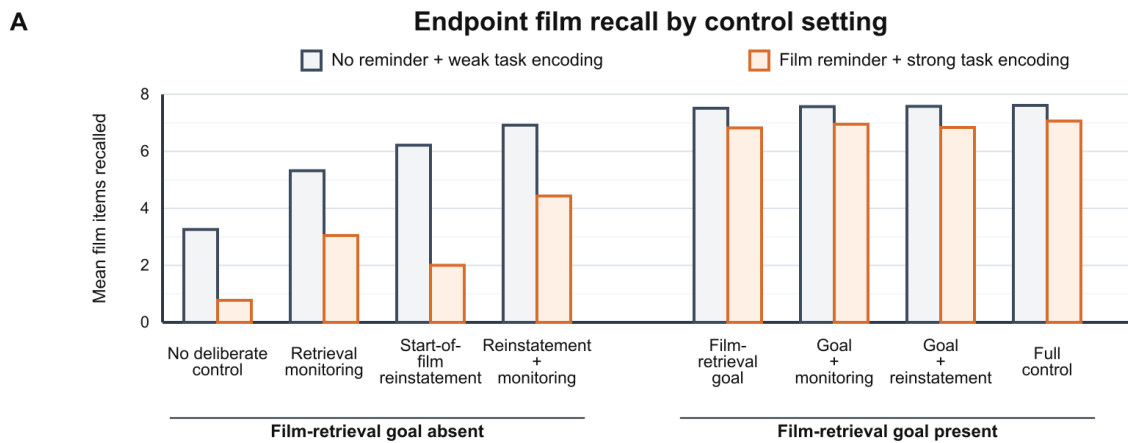


Figure 9 shows that endpoint loss remains large when the film-retrieval goal is absent. Start-of-film reinstatement and retrieval monitoring increase film recall in some conditions, yet the smallest endpoint differences occur in the goal-present conditions. The key protection comes from film-directed support that task competitors do not receive.

Figure 10***Simulation 2: a film-retrieval goal separates film-directed retrieval from task context overlap.***

The high-interference condition was held fixed: a film reminder preceded strong task encoding, and weak film cues were supplied periodically during recall. A film-retrieval goal is an internally maintained retrieval state that supports film items during deliberate recall, distinct from the external film reminder and test-phase film cues. (A) Mean overlap between task-encoding contexts and film-directed retrieval cues, averaging across encoded task positions for each encoded film position. (B) Full overlap matrices for the same condition. The left matrix uses temporal context alone; the right matrix adds the film-retrieval goal to the film-directed retrieval cue. Brighter cells indicate greater overlap with task-encoding context. Adding the film-retrieval goal reduces overlap between film-directed retrieval cues and task-encoding contexts, providing a route to film recall that depends less on temporal context remaining selective for film items.

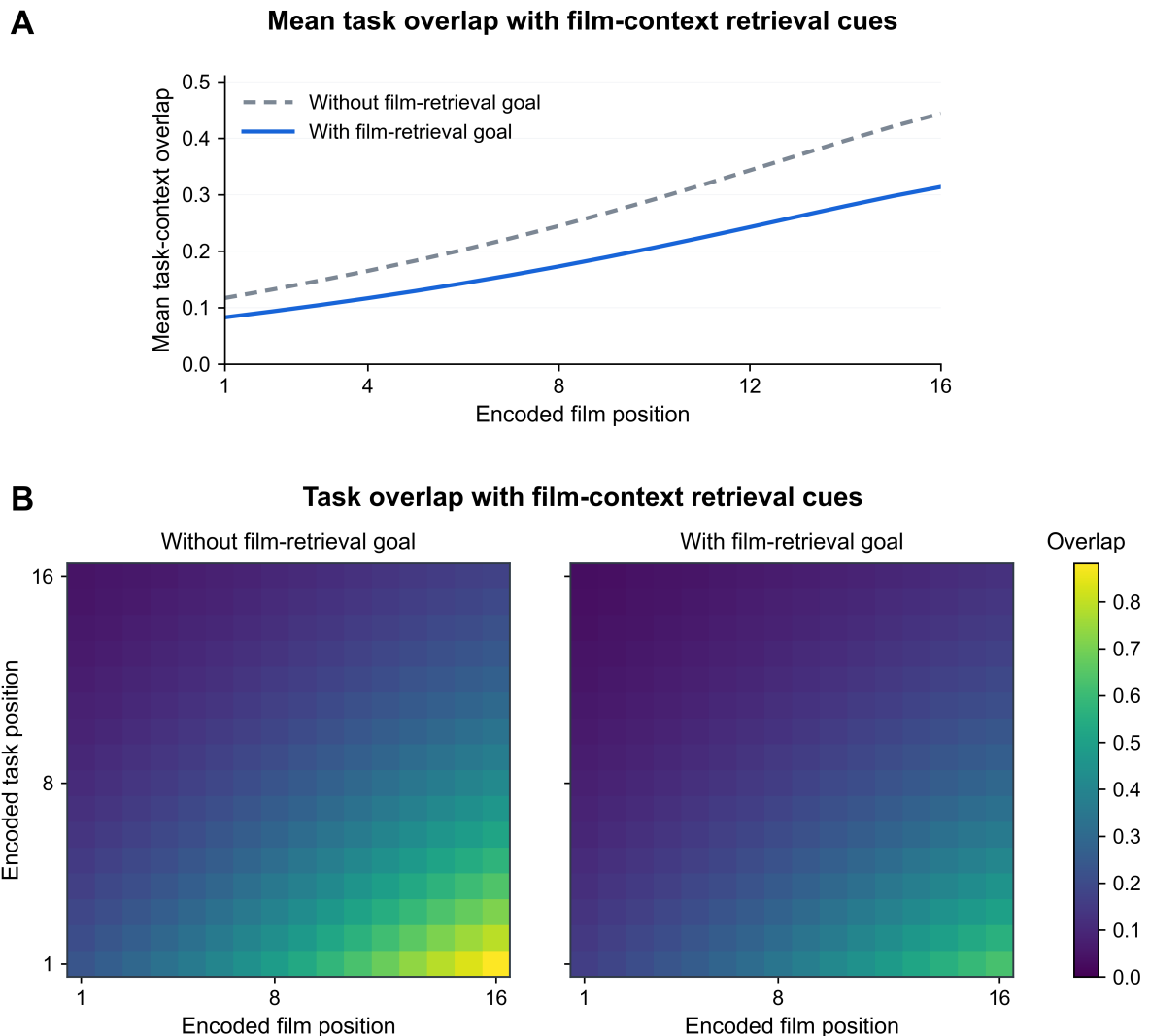
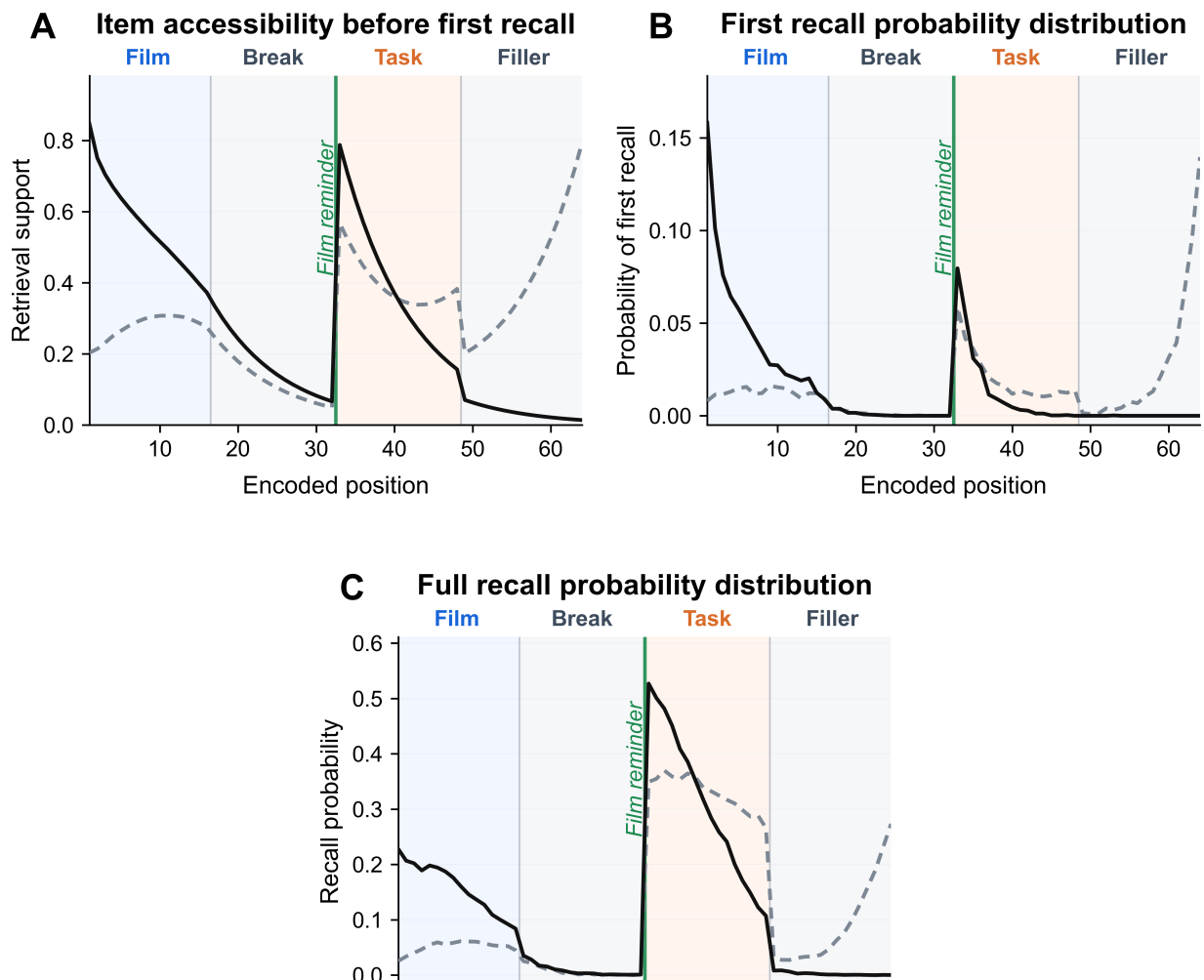


Figure Figure 10 explains why the goal-present conditions are less affected by reminder-linked task encoding. Adding the goal lowers task overlap with film-directed retrieval cues, so deliberate recall can favor film items without relying as strongly on temporal context that also supports task competitors.

The remaining diagnostics show how the other controls help without providing the same broad protection. Start-of-film context reinstatement changes where search begins, so its benefits are concentrated around early film positions.

Figure 11***Simulation 2: start-of-film context reinstatement protects early encoded film positions.***

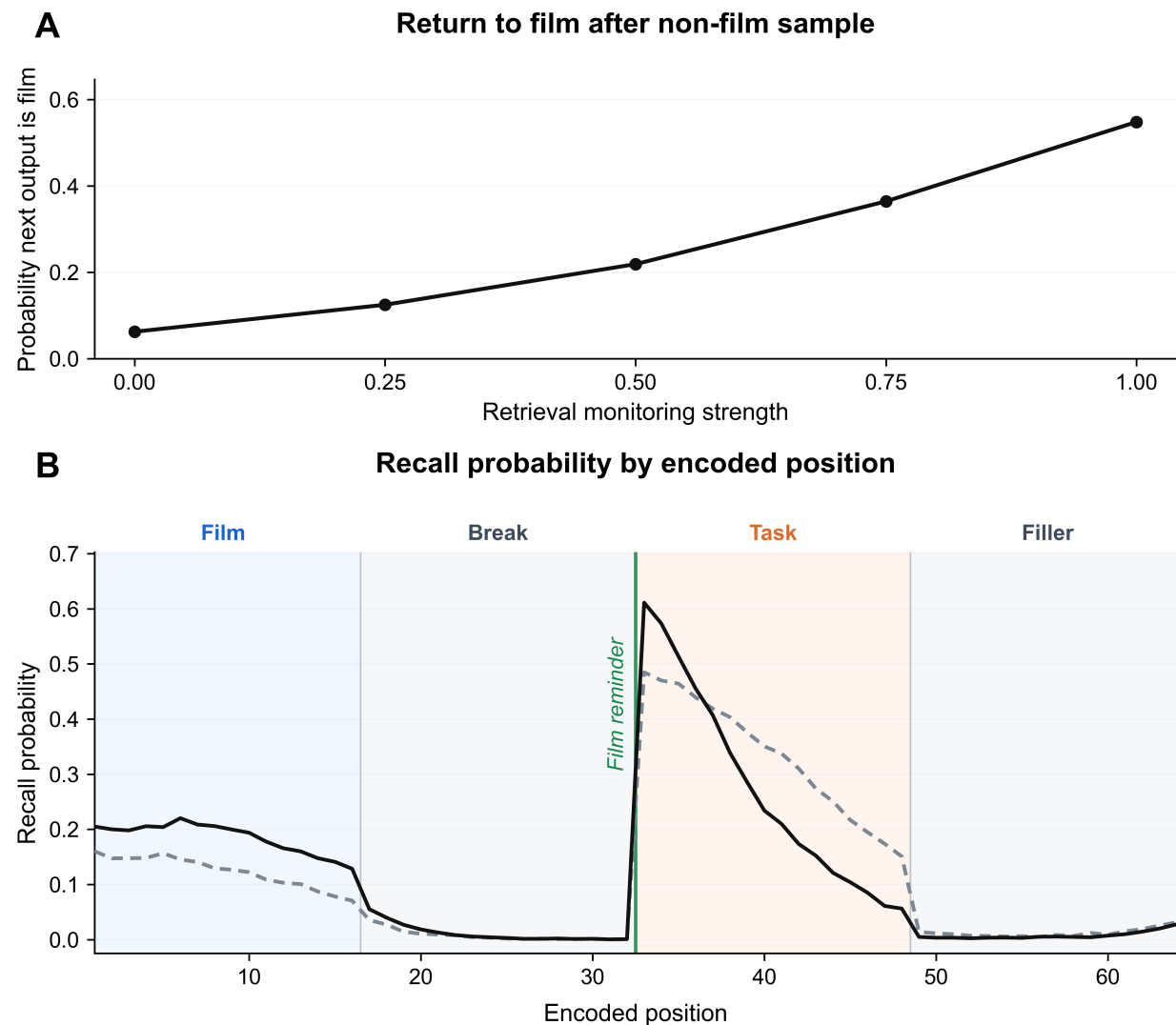
Unguided recall was compared with start-of-film context reinstatement while the condition with a pre-task film reminder and strong task encoding was held fixed and retrieval monitoring was disabled. Start-of-film context reinstatement initializes retrieval by drifting the current context toward the state present at the start of film encoding. Weak film cues were supplied periodically during recall. Shaded bands mark film, break, task, and filler positions; the green line marks the reminder immediately before task encoding. (A) Item accessibility before the first recall attempt. Reinstatement increases accessibility most strongly for early encoded film positions. (B) First recall probability distribution. Reinstatement shifts recall initiation toward early film positions. (C) Full recall probability distribution. The early-film bias remains visible across the recall period, while later film positions remain more vulnerable under reminder-linked strong task encoding. Start-of-film context reinstatement therefore protects early encoded film positions by shifting retrieval initiation toward the start of the film sequence.



Retrieval monitoring addresses a later part of the retrieval sequence. When search samples a non-film item, monitoring limits how strongly that sample redirects subsequent retrieval.

Figure 12

Simulation 2: retrieval monitoring supports return to film recall. The condition with a pre-task film reminder and strong task encoding was held fixed, start-of-film context reinstatement was present, and retrieval monitoring strength was varied. Retrieval monitoring dampens context updating after non-film samples during deliberate recall. Weak film cues were supplied periodically during recall. **(A)** Model-diagnostic probability that a non-film sample was followed by a film output as retrieval monitoring increased. **(B)** Recall probability by encoded position with no retrieval monitoring versus the moderate retrieval-monitoring setting used in the Simulation 2 anchor figure. Shaded bands mark film, break, task, and filler positions; the green line marks the reminder immediately before task encoding. Retrieval monitoring therefore supports film recall after search is pulled away from the film sequence, increasing film recall probability under reminder-linked strong task encoding.



Recognition-Like Test Format

Recognition tests in the trauma-film literature present candidate film stills or statements and ask participants to judge whether they appeared in the film ([Lau-Zhu et al., 2021](#)). Retrieved-context models have also been extended to recognition more generally: a test probe can reinstate its associated context, and the match between reinstated context and the current retrieval context can contribute to an old-new decision ([Healey & Kahana, 2016](#); [Kahana, 2020](#)). Related recognition findings show that temporal context retrieved by one probe can influence recognition of nearby studied probes ([Schwartz et al., 2005](#)). Here, the model isolates the access step that recognition provides. The candidate film item is available first and can reinstate associated context before any need for that item to win a context-to-item recall competition. This analysis does not model decision criteria, confidence, or complete foil processing. It asks whether presenting the candidate item changes where reminder-linked task competition enters retrieval.

Figure 13

Item-to-context retrieval bypasses reminder-linked task interference. (A) CMR stores bidirectional associations between item features and context. Feature-to-context memory lets an available item reinstate its associated context, whereas context-to-feature memory lets context cue candidate items. (B) When film-associated context cues items, film items and reminder-linked task items can both receive support. When a film item reinstates context, retrieval targets context units instead, so task items are not competitors in that step. (C) In recognition, the tested film item is already available. The item retrieves associated context, which is compared with ongoing test context; stronger matches provide stronger evidence that the item appeared in the film. (D) Film recognition accuracy is plotted across the same reminder and task-encoding conditions used in the recall simulations. Accuracy is hit rate minus false-alarm rate for unstudied film-like lures. The reminder-plus-strong-task condition does not produce a corresponding reduction in recognition accuracy.

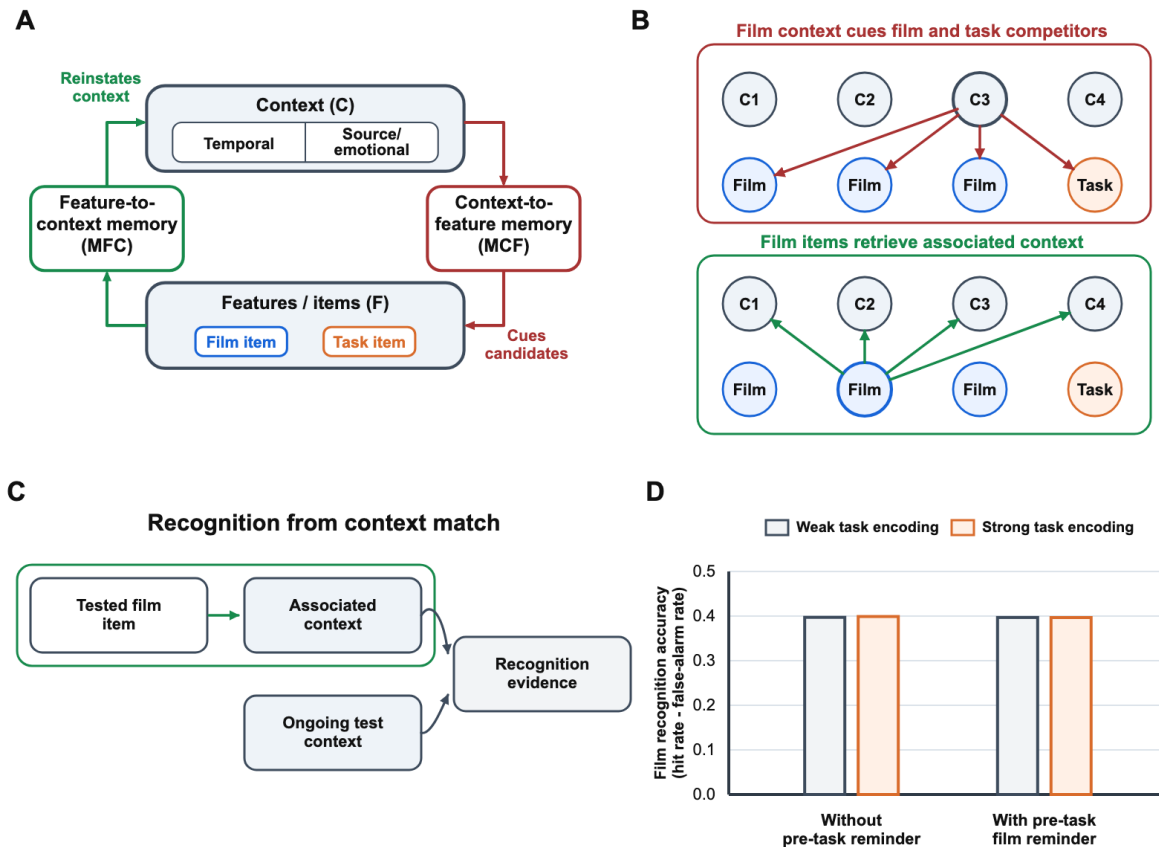


Figure 13 shows the resulting test-format consequence. In context-guided recall, reminder-linked task items can compete when the current context supports both film and task candidates. In recognition-like tests, the presented candidate item reinstates context directly. The task items created by reminder-linked encoding therefore have less opportunity to compete at the initial access step, even though the same learned associations are present in memory.

General Discussion

References

- Asselbergs, J., Riper, H., Engelhard, I. M., Mannes, F., & Sijbrandij, M. (2024). The effectiveness of two novel approaches to prevent intrusions: A pilot study comparing Tetris_dualtask and imagery rescripting to control. *Journal of Behavior Therapy and Experimental Psychiatry*, 82, 101920. <https://doi.org/10.1016/j.jbtep.2023.101920>
- Asselbergs, J., van Bentum, J., Riper, H., Cuijpers, P., Holmes, E. A., & Sijbrandij, M. (2023). A systematic review and meta-analysis of the effect of cognitive interventions to prevent intrusive memories using the trauma film paradigm. *Journal of Psychiatric Research*, 159, 116–129. <https://doi.org/10.1016/j.jpsychires.2023.01.028>
- Baddeley, A. D., & Andrade, J. (2000). Working memory and the vividness of imagery. *Journal of Experimental Psychology: General*, 129(1), 126–145. <https://doi.org/10.1037/0096-3445.129.1.126>
- Beckenstrom, A. C., Bonsall, M. B., Markham, A., Slade, O., Ramineni, V., Iyadurai, L., Islam, Z., Highfield, J., Jaki, T., Goodwin, G. M., Dias, R., Daniels, R., Malik, A., Summers, C., Kingslake, J., & Holmes, E. A. (2026). A digital imagery-competing task intervention for stopping intrusive memories in trauma-exposed health-care staff during the COVID-19 pandemic in the UK: A bayesian adaptive randomised clinical trial. *The Lancet Psychiatry*, 13(3), 233–247. [https://doi.org/10.1016/s2215-0366\(25\)00397-9](https://doi.org/10.1016/s2215-0366(25)00397-9)
- Brewin, C. R. (2014). Episodic memory, perceptual memory, and their interaction: Foundations for a theory of posttraumatic stress disorder. *Psychological Bulletin*, 140(1), 69–97. <https://doi.org/10.1037/a0033722>

Brewin, C. R., Dalgleish, T., & Joseph, S. (1996). A dual representation theory of posttraumatic stress disorder. *Psychological Review*, 103(4), 670–686.

<https://doi.org/10.1037/0033-295X.103.4.670>

Brewin, C. R., Gregory, J. D., Lipton, M., & Burgess, N. (2010). Intrusive images in psychological disorders: Characteristics, neural mechanisms, and treatment implications.

Psychological Review, 117(1), 210–232. <https://doi.org/10.1037/a0018113>

Bussemeyer, J. R., & Wang, Y.-M. (2000). Model comparisons and model selections based on generalization criterion methodology. *Journal of Mathematical Psychology*, 44(1), 171–189.

<https://doi.org/10.1006/jmps.1999.1282>

Cohen, R. T., & Kahana, M. J. (2022). A memory-based theory of emotional disorders.

Psychological Review, 129(4), 742–776. <https://doi.org/10.1037/rev0000334>

Deepröse, C., Zhang, S., DeJong, H., Dalgleish, T., & Holmes, E. A. (2012). Imagery in the aftermath of viewing a traumatic film: Using cognitive tasks to modulate the development of involuntary memory. *Journal of Behavior Therapy and Experimental Psychiatry*, 43(2),

758–764. <https://doi.org/10.1016/j.jbtep.2011.10.008>

Dupertuys, J., Shulhan, A., & Talmi, D. (2026). *Ukraine Memory experiment data set*.

Ehlers, A., & Clark, D. M. (2000). A cognitive model of posttraumatic stress disorder. *Behaviour Research and Therapy*, 38(4), 319–345. [https://doi.org/10.1016/S0005-7967\(99\)00123-0](https://doi.org/10.1016/S0005-7967(99)00123-0)

Healey, M. K., & Kahana, M. J. (2014). Is memory search governed by universal principles or idiosyncratic strategies? *Journal of Experimental Psychology: General*, 143(2), 575–596.

<https://doi.org/10.1037/a0033715>

Healey, M. K., & Kahana, M. J. (2016). A four-component model of age-related memory change.

Psychological Review, 123(1), 23–69. <https://doi.org/10.1037/rev0000015>

Healey, M. K., & Wahlheim, C. N. (2024). PEPPR: A post-encoding pre-production reinstatement model of dual-list free recall. *Memory & Cognition*, 52(1), 163–181.

<https://doi.org/10.3758/s13421-023-01453-z>

Hemi, A., Sopp, M. R., Perel, A., Holmes, E. A., & Levy-Gigi, E. (2023). Cognitive flexibility

moderates the efficacy of a visuospatial intervention following exposure to analog trauma.

Journal of Behavior Therapy and Experimental Psychiatry, 81, 101858.

<https://doi.org/10.1016/j.jbtep.2023.101858>

Herrema, H. G., & Kahana, M. J. (2024). *First recall costs and benefits*. PsyArXiv.

<https://doi.org/10.31234/osf.io/sm3fw>

Holmes, E. A., & Bourne, C. (2008). Inducing and modulating intrusive emotional memories: A review of the trauma film paradigm. *Acta Psychologica*, 127(3), 553–566.

<https://doi.org/10.1016/j.actpsy.2007.11.002>

Holmes, E. A., Brewin, C. R., & Hennessey, R. G. (2004). Trauma films, information processing, and intrusive memory development. *Journal of Experimental Psychology: General*, 133(1), 3–22. <https://doi.org/10.1037/0096-3445.133.1.3>

Holmes, E. A., James, E. L., Coode-Bate, T., & Deeptose, C. (2009). Can playing the computer game Tetris reduce the build-up of flashbacks for trauma? A proposal from cognitive science. *PLoS ONE*, 4(1), e4153. <https://doi.org/10.1371/journal.pone.0004153>

Holmes, E. A., James, E. L., Kilford, E. J., & Deeptose, C. (2010). Key steps in developing a cognitive vaccine against traumatic flashbacks: Visuospatial Tetris versus verbal Pub Quiz. *PLoS ONE*, 5(11), e13706. <https://doi.org/10.1371/journal.pone.0013706>

Horsch, A., Vial, Y., Favrod, C., Harari, M. M., Blackwell, S. E., Watson, P., Iyadurai, L., Bonsall, M. B., & Holmes, E. A. (2017). Reducing intrusive traumatic memories after emergency caesarean section: A proof-of-principle randomized controlled study. *Behaviour Research and Therapy*, 94, 36–47. <https://doi.org/10.1016/j.brat.2017.03.018>

Howard, M. W., & Kahana, M. J. (1999). Contextual variability and serial position effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25(4), 923–941. <https://doi.org/10.1037/0278-7393.25.4.923>

Howard, M. W., & Kahana, M. J. (2002). A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46(3), 269–299.

<https://doi.org/10.1006/jmps.2001.1388>

Howard, M. W., Youker, T. E., & Venkatadass, V. S. (2008). The persistence of memory:

Contiguity effects across hundreds of seconds. *Psychonomic Bulletin & Review*, 15(1), 58–63.

Hupbach, A., Hardt, O., Gomez, R., & Nadel, L. (2008). The dynamics of memory:

Context-dependent updating. *Learning & Memory*, 15(8), 574–579.

<https://doi.org/10.1101/lm.1022308>

Iyadurai, L., Blackwell, S. E., Meiser-Stedman, R., Watson, P. C., Bonsall, M. B., Geddes, J. R.,

Nobre, A. C., & Holmes, E. A. (2018). Preventing intrusive memories after trauma via a brief intervention involving Tetris computer game play in the emergency department: A proof-of-concept randomized controlled trial. *Molecular Psychiatry*, 23(3), 674–682.

<https://doi.org/10.1038/mp.2017.23>

Iyadurai, L., Highfield, J., Kanstrup, M., Markham, A., Ramineni, V., Guo, B., Jaki, T.,

Kingslake, J., Goodwin, G. M., Summers, C., Bonsall, M. B., & Holmes, E. A. (2023).

Reducing intrusive memories after trauma via an imagery-competing task intervention in COVID-19 intensive care staff: A randomised controlled trial. *Translational Psychiatry*, 13(1), 290. <https://doi.org/10.1038/s41398-023-02578-0>

Iyadurai, L., Visser, R. M., Lau-Zhu, A., Porcheret, K., Horsch, A., Holmes, E. A., & James, E. L.

(2019). Intrusive memories of trauma: A target for research bridging cognitive science and its clinical application. *Clinical Psychology Review*, 69, 67–82.

<https://doi.org/10.1016/j.cpr.2018.08.005>

James, E. L., Bonsall, M. B., Hoppitt, L., Tunbridge, E. M., Geddes, J. R., Milton, A. L., &

Holmes, E. A. (2015). Computer game play reduces intrusive memories of experimental trauma via reconsolidation-update mechanisms. *Psychological Science*, 26(8), 1201–1215.

<https://doi.org/10.1177/0956797615583071>

James, E. L., Lau-Zhu, A., Clark, I. A., Visser, R. M., Hageraars, M. A., & Holmes, E. A. (2016).

The trauma film paradigm as an experimental psychopathology model of psychological trauma: Intrusive memories and beyond. *Clinical Psychology Review*, 47, 106–142.

<https://doi.org/10.1016/j.cpr.2016.04.010>

- Kahana, M. J. (1996). Associative retrieval processes in free recall. *Memory & Cognition*, 24(1), 103–109.
- Kahana, M. J. (2020). Computational models of memory search. *Annual Review of Psychology*, 71, 107–138. <https://doi.org/10.1146/annurev-psych-010418-103358>
- Kanstrup, M., Kontio, E., Geranmayeh, A., Olofsdotter Lauri, K., Moulds, M. L., & Holmes, E. A. (2021). A single case series using visuospatial task interference to reduce the number of visual intrusive memories of trauma with refugees. *Clinical Psychology & Psychotherapy*, 28(1), 109–123. <https://doi.org/10.1002/cpp.2489>
- Kanstrup, M., Singh, L., Leehr, E. J., Göransson, K. E., Pihlgren, S. A., Iyadurai, L., Dahl, O., Falk, A.-C., Lindström, V., Hadziosmanovic, N., Gabrysch, K., Moulds, M. L., & Holmes, E. A. (2024). A guided single session intervention to reduce intrusive memories of work-related trauma: A randomised controlled trial with healthcare workers in the COVID-19 pandemic. *BMC Medicine*, 22(1), 403. <https://doi.org/10.1186/s12916-024-03569-8>
- Kessler, H., Schmidt, A.-C., James, E. L., Blackwell, S. E., von Rauchhaupt, M., Harren, K., Kehyayan, A., Clark, I. A., Sauvage, M., Herpertz, S., Axmacher, N., & Holmes, E. A. (2020). Visuospatial computer game play after memory reminder delivered three days after a traumatic film reduces the number of intrusive memories of the experimental trauma. *Journal of Behavior Therapy and Experimental Psychiatry*, 67, 101454. <https://doi.org/10.1016/j.jbtep.2019.01.006>
- Kindt, M., Soeter, M., & Vervliet, B. (2009). Beyond extinction: Erasing human fear responses and preventing the return of fear. *Nature Neuroscience*, 12(3), 256–258. <https://doi.org/10.1038/nn.2271>
- Kragel, J. E., Morton, N. W., & Polyn, S. M. (2015). Neural activity in the medial temporal lobe reveals the fidelity of mental time travel. *Journal of Neuroscience*, 35(7), 2914–2926. <https://doi.org/10.1523/JNEUROSCI.3378-14.2015>
- Krans, J., Näring, G., Becker, E. S., & Holmes, E. A. (2009). Intrusive trauma memory: A review and functional analysis. *Applied Cognitive Psychology*, 23(8), 1076–1088.

<https://doi.org/10.1002/acp.1611>

- Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2019). Intrusive memories and voluntary memory of a trauma film: Differential effects of a cognitive interference task after encoding. *Journal of Experimental Psychology: General*, 148(12), 2154–2180. <https://doi.org/10.1037/xge0000598>
- Lau-Zhu, A., Henson, R. N., & Holmes, E. A. (2021). Selectively interfering with intrusive but not voluntary memories of a trauma film: Accounting for the role of associative memory. *Clinical Psychological Science*, 9(6), 1128–1143. <https://doi.org/10.1177/2167702621998315>
- Lohnas, L. J., Healey, M. K., & Davachi, L. (2023). Neural temporal context reinstatement of event structure during memory recall. *Journal of Experimental Psychology: General*, 152(7), 1840–1872. <https://doi.org/10.1037/xge0001354>
- Lohnas, L. J., Polyn, S. M., & Kahana, M. J. (2015). Expanding the scope of memory search: Modeling intralist and interlist effects in free recall. *Psychological Review*, 122(2), 337–363. <https://doi.org/10.1037/a0039036>
- McConnell, T. G. W., Bernhard, H., Holmes, E. A., Henson, R., & Talmi, D. (2026). *Testing the dual representation account of memory for trauma*. PsyArXiv preprint. https://osf.io/preprints/psyarxiv/m3pnz_v1/
- Morton, N. W., & Polyn, S. M. (2016). A predictive framework for evaluating models of semantic organization in free recall. *Journal of Memory and Language*, 86, 119–140. <https://doi.org/10.1016/j.jml.2015.10.002>
- Polyn, S. M., Kragel, J. E., Morton, N. W., McCluey, J. D., & Cohen, Z. D. (2012). The neural dynamics of task context in free recall. *Neuropsychologia*, 50(4), 447–457. <https://doi.org/10.1016/j.neuropsychologia.2011.08.025>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009a). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review*, 116(1), 129–156. <https://doi.org/10.1037/a0014420>
- Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009b). Task context and organization in free recall. *Neuropsychologia*, 47(11), 2158–2163.

<https://doi.org/10.1016/j.neuropsychologia.2009.02.013>

Ramineni, V., Millroth, P., Iyadurai, L., Jaki, T., Kingslake, J., Highfield, J., Summers, C.,

Bonsall, M. B., & Holmes, E. A. (2023). Treating intrusive memories after trauma in healthcare workers: A bayesian adaptive randomised trial developing an imagery-competing task intervention. *Molecular Psychiatry*, 28(7), 2985–2994.

<https://doi.org/10.1038/s41380-023-02062-7>

Schwartz, G., Howard, M. W., Jing, B., & Kahana, M. J. (2005). Shadows of the past: Temporal retrieval effects in recognition memory. *Psychological Science*, 16(11), 898–904.

<https://doi.org/10.1111/j.1467-9280.2005.01634.x>

Sederberg, P. B., Gershman, S. J., Polyn, S. M., & Norman, K. A. (2011). Human memory reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin & Review*, 18(3), 455–468. <https://doi.org/10.3758/s13423-011-0086-9>

Sederberg, P. B., Howard, M. W., & Kahana, M. J. (2008). A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4), 893–912.

<https://doi.org/10.1037/a0013396>

Talmi, D. (2013). Enhanced emotional memory: Cognitive and neural mechanisms. *Current Directions in Psychological Science*, 22(6), 430–436.

<https://doi.org/10.1177/0963721413498893>

Talmi, D., Lohnas, L. J., & Daw, N. D. (2019). A retrieved context model of the emotional modulation of memory. *Psychological Review*, 126(4), 455–485.

<https://doi.org/10.1037/rev0000132>

Talmi, D., Luk, B. T. C., McGarry, L. M., & Moscovitch, M. (2007). The contribution of relatedness and distinctiveness to emotionally-enhanced memory. *Journal of Memory and Language*, 56(4), 555–574. <https://doi.org/10.1016/j.jml.2007.01.002>

Talmi, D., & Moscovitch, M. (2004). Can semantic relatedness explain the enhancement of memory for emotional words? *Memory & Cognition*, 32(5), 742–751.

<https://doi.org/10.3758/bf03195864>

- Varma, M. M., Zeng, S., Singh, L., Holmes, E. A., Huang, J., Chiu, M. H., & Hu, X. (2024). A systematic review and meta-analysis of experimental methods for modulating intrusive memories following lab-analogue trauma exposure in non-clinical populations. *Nature Human Behaviour*, 8(10), 1968–1987. <https://doi.org/10.1038/s41562-024-01956-y>
- Wessel, I., Krans, J., Albers, C. J., Chauhan, N., Ferreira de Sá, D. S., Hauck, A., Jaswetz, L., Michael, T., Nixon, R. D. V., Pearson, D. G., Slattery, I., Smeets, T., Takarangi, M. K. T., Willems, T. R. A., & van Schie, K. (2025). Evidence that Tetris reduces immediate but not subsequent daily intrusions of a trauma film: A multilab replication study. *Collabra: Psychology*, 11(1), 130791. <https://doi.org/10.1525/collabra.130791>
- Zaromb, F. M., Howard, M. W., Dolan, E. D., Sirotin, Y. B., Tully, M., Wingfield, A., & Kahana, M. J. (2006). Temporal associations and prior-list intrusions in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 32(4), 792–804. <https://doi.org/10.1037/0278-7393.32.4.792>
- Zhou, C. Y., Talmi, D., Daw, N. D., & Mattar, M. G. (2025). Episodic retrieval for model-based evaluation in sequential decision tasks. *Psychological Review*, 132(1), 18–49. <https://doi.org/10.1037/rev0000505>

Table 1

Model parameters and roles in the model specification. Base parameters are the unscaled values used by the model equations before simulation-specific multipliers are applied. A phase is a segment of the simulated paradigm, such as film encoding, intervening delay, reminder, task encoding, post-task delay, or retrieval.

Symbol	Name	Role
β_{enc}	encoding drift rate	Context drift during study
β_{start}	start drift rate	Drift toward start-of-film context at recall onset
β_{rec}	recall drift rate	Context drift after recall
γ	feature-to-context learning rate	Learning rate for M^{FC}
λ	context-to-feature learning scale	Baseline or phase-scaled learning rate for M^{CF}
α	shared support	Pre-experimental shared support in M^{CF}
δ	self support	Pre-experimental self-support in M^{CF}
ϕ_s	primacy scale	Initial boost to context-to-feature learning
ϕ_d	primacy decay	Decay rate of primacy boost
τ	choice sensitivity	Sensitivity of choice to support differences
β_{emot}	source-context drift rate	Source context drift
ϕ_{emot}	source learning scale	Learning scale for the marked source feature
η_s	source learning baseline	Baseline source-context learning for neutral/default items
$\beta_{film-src}$	film-source orientation	Fixed partial reinstatement of marked film-source context at

Table 2

Overview of what is held fixed, varied, and measured across simulations.

Simulation	Main fixed structure	Varied or compared	Primary readout
1	Unguided retrieval settings; film, delay, task, and filler phases	No-reminder versus with-reminder task encoding; task encoding multiplier	Film and task recall by encoded position; film-task context overlap
2	Reminder condition and task encoding strength crossed as in Simulation 1	Unguided retrieval versus deliberate recall; start-of-film context reinstatement, maintained film-retrieval goal, and retrieval monitoring	Film recall across retrieval modes; endpoint differences across control settings; recall initiation and film return after off-target samples
Recognition	Same reminder and task-encoding conditions used in recall simulations, without a full recognition-decision model	Context-to-item recall versus candidate film items presented for judgment	Whether item-to-context access changes where interference enters retrieval

Table 3

Default numerical parameter values. Values are from a pooled eCMR fit to the immediate free-recall sequences from Dupertuys et al. (2026), with α fixed to zero, except for the source-context and primacy settings described in the text.

Symbol	Value	Role
β_{enc}	0.440	Encoding drift rate
β_{start}	0.766	Start drift rate
β_{rec}	0.678	Recall drift rate
γ	0.468	Feature-to-context learning rate
δ	0.00022	Pre-experimental self-support
α	0	Pre-experimental shared support
ϕ_s	0.100	Primacy scale used in simulations
ϕ_d	1.500	Primacy decay used in simulations
τ	3.562	Choice sensitivity
β_{emot}	0.696	Source-context drift rate
ϕ_{emot}	0.130	Learning scale for the marked source feature
η_s	0.050	Baseline source-context-to-feature learning
$\beta_{film-src}$	0.500	Fixed film-source reinstatement at retrieval onset
θ_s	0.0123	Stop scale
θ_r	0.372	Stop growth

Table 4

Phase structure and settings for Simulation 1. Scale values multiply the corresponding base rates or learning strengths.

Phase or setting	Current value	Role
Film items	16	Target episode
Break/buffer items	16	Intervening non-film experience before reminder/task
Film reminder cues	16	Sequential film cues, no new learning
Task items	16	Potential competitors
Filler items	16	Post-task non-film experience
Maximum sampling attempts	48	Upper bound on sampled retrieval attempts
Simulated trials per condition	5000	For each reminder condition and task encoding multiplier
Break and filler drift/learning scales	1.0	Baseline non-film encoding settings
No-reminder drift scales	0.0	Omits reminder-start and reminder-cue reinstatement
With-reminder drift scales	1.0	Applies reminder-start and film cue reinstatement
Task encoding multiplier	1.0 or 2.0	Multiplies task-phase context-to-feature learning during task encoding
Film source marking	On	Film items carry the marked source feature
Task source marking	Off	Task items use the